

bibliosense

Bibliometric Analysis Report — AI Interpreted

Non-Newtonian Blood Rheology and Computational Hemodynamics 2016– 2025

Mapping viscoelastic fluid flow, coronary artery hemodynamics, and hemorheology across a decade of research.

697 records · 2016–2025 · — themes · strategy: auto

Generated by bibliosense · claude_cli/claude-sonnet-5 · 03 July 2026

Key findings at a glance

RECORDS	PERIOD	GROWTH (CAGR)	THEMES
697	2016–2025 (10 yrs)	+6.4%/yr	—
H-INDEX	G-INDEX	M-INDEX	TOTAL CITATIONS
49	74	4.90	11,587
AUTHORS	COLLABORATION INDEX	MULTI-AUTHOR %	TOP VENUE
2,634	4.83	95%	Physics of Fluids

Most-cited paper: “Blood rheology: Key parameters, impact on blood flow, role in sickle cell disease and effects of exercise” — 429 citations

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⚠ FULL AI INTERPRETED REPORT — Interpretive narrative sections in this document were generated by **bibliosense[ai] · claude_cli · claude-sonnet-5**. They are based on pattern recognition applied to the bibliometric data and may contain errors, oversimplifications, or hallucinations. Verify all AI-generated insights against the data tables before citing or sharing.

0. Information sources

The analysis was run on a single dataset; the database + query metadata below documents the source for reproducibility.

Database	Query date	Raw records	After dedup
Scopus	2026-07-03	697	697

Verbatim search queries

Scopus · queried 2026-07-03

TITLE(("non-Newtonian fluid*" OR "non Newtonian fluid*" OR "shear-thinning" OR "shear thinning" OR "viscoelastic*" OR "rheolog*")) AND TITLE-ABS-KEY(("biological system*" OR "blood flow" OR "biofluid*" OR "bio-fluid*" OR "hemodynamic*" OR "haemodynamic*" OR "mucus" OR "synovial fluid")) AND PUBYEAR > 2015 AND PUBYEAR < 2026 AND LANGUAGE(english)

Search strategy development

Original idea (as expressed by the researcher): “*what are the latest advances in non newtonian fluids in biological systems*”

Candidate research questions proposed by the assistant:

1. What is the current state of research on non-Newtonian fluid behavior in biological systems, and how have key application areas evolved over the past decade? ← **selected**
2. To what extent do non-Newtonian rheological models (e.g., Carreau, power-law, Casson) improve the accuracy of blood flow simulations compared to Newtonian models in cardiovascular studies?
3. What advances have been made in experimental and computational methods for characterizing non-Newtonian behavior of biological fluids (e.g., mucus, synovial fluid, blood) at the microscale, and how are these methods being applied in biomedical engineering or microfluidics?

Selected research question: *What is the current state of research on non-Newtonian fluid behavior in biological systems, and how have key application areas evolved over the past decade?*

Document-type composition (AI-classified)

Type	Records	Share
Review article	9	1%
Editorial / commentary	9	1%
Other	679	97%

This corpus is dominated by a data-quality gap rather than a genuine publication-type skew: 78 of 80 records (97.5%) have empty venue strings and were bucketed 'Other', far exceeding any of the named significance thresholds and signaling that venue metadata must be recovered before meaningful classification is possible. Only two records yielded title-based cues — one Editorial/commentary and one Review article (1.25% each) — both well below the 'Heavy review share' or other consolidation thresholds, so no substantive maturity or venue-mix signal can be drawn from this sample as given.

Classification based on a sample of 80 records (title + venue), scaled to the 697-record corpus. Edge cases default to 'Other'.

1. Executive summary

This review covers 697 records published between 2016 and 2025. United States leads national contributions with 158 records. The keyword co-occurrence network resolves into 7 thematic clusters. The corpus has an h-index of 49 and a g-index of 74. A Bradford core of 32 venues concentrates 232 of the publications. The most prominent recent emerging theme is 'magnetic field' in 2022–2024. Between 2022–2024 and 2025, the clearest thematic shift is toward 'mechanical', whose share of period keywords rose from 2.2% to 4.9%.

1.1 AI insights of research opportunities

The keyword co-occurrence network resolved into 7 thematic sub-fields, including 'Human Blood Rheology and Hemorrheology', 'Viscoelastic Fluid Flow Dynamics', 'Non-Newtonian Blood Flow Haemodynamics'. The full thematic breakdown appears in §11.1.

The corpus has solid coverage of rheological and viscosity-focused characterization of biological fluids is a core strength: this theme dominates the top-keyword list (rheology, viscosity, non-newtonian fluids) across the full 2016–2025 span of the 697-record corpus and 388 distinct publication sources.; hemodynamics and blood-flow viscoelasticity form a second deeply anchored theme, co-occurring with 'blood flow', 'hemodynamics', and 'viscoelastic properties' among the top 15 keywords spanning the same decade (2016–2025) and broad venue base of 388 sources.. Coverage strengths and limitations are detailed in §1.2.

bibliosense[ai] identified 6 corpus-specific research observations — including topic concentration patterns and temporal coverage shifts. Full details appear in §6.4.

1.2 Insights into potential AI-identified research gaps

Strictly informed by the bibliometric analysis above (top keywords, cluster labels, emerging themes, time trends, co-occurrence patterns), the AI assistant has surfaced the following research gaps and candidate research directions. Each gap names the specific corpus signal that motivates it. Use them to seed a Literature Review — the toolbar's “Continue to Literature Review” action carries these gaps and their candidate questions across automatically.

G-1: Biomaterial Microrheology Cluster Disconnect

The seven named clusters in this corpus (e.g., 'Human Blood Rheology and Hemorrheology', 'Computational Hemodynamics In Coronary Arteries') are built almost entirely around blood and vascular flow, yet a substantial share of the 2025 top-30 records address soft biomaterial and tissue microrheology unrelated to blood: AFM nanoindentation for 'predicting nanoscale stiffness in soft biomaterials' (R-1), microfluidic multiple-particle-tracking microrheology of biogels (R-13), tensile acoustic rheometry of soft viscoelastic biomaterials (R-21), and photothermal AFM nanorheology of polymeric cell substrates (R-27). None of these records show a hemorrheology-specific cluster label; they sit outside the named taxonomy, indicating the cluster structure has not caught up with this emerging non-blood biomaterials characterization subfield.

Why: This is a bridge gap because the microrheology/biomaterials keyword set (AFM, acoustic rheometry, microfluidic chips) co-occurs internally but rarely connects to the blood/hemodynamics keywords that dominate the seven established clusters.

Anchored in corpus papers:

- *Viscoelastic interpretation of AFM nanoindentation for predicting nanoscale stiffness in soft biomaterials* (2025) — Non-blood biomaterial viscoelasticity method absent from named clusters
- *Microrheology-on-a-chip: A microfluidic pathway to measure the viscoelasticity of biogels* (2025) — Microfluidic microrheology unconnected to hemorheology cluster labels

Candidate research questions:

- What measurement techniques (AFM nanoindentation, acoustic rheometry, microfluidic microrheology) are used to characterize soft biomaterial viscoelasticity, and how do their reported outputs compare across biomaterial types? — *Maps the fragmented technique landscape before comparative synthesis is possible.*

Specific to this corpus: Targets the unclustered 2025 biomaterials records (R-1, R-13, R-21, R-27) sitting outside the seven named clusters.

Suggested review type: scoping

- Can nanoscale/microscale viscoelastic characterization methods developed for soft biomaterials be adapted to characterize blood or vascular tissue rheology? — *Tests transferability of methods across the corpus's disconnected subfields.*

Specific to this corpus: Bridges the biomaterials records with the 'Human Blood Rheology and Hemorheology' and 'Non-Newtonian Blood Flow Haemodynamics' clusters.

Suggested review type: systematic

G-2: Mucus Rheology Framing Gap

Multiple 2025 records center on mucus and mucus-analog viscoelasticity - thixo-viscoelastoplastic BMP+ modeling of 'human mucus-sputum' in healthy and Cystic Fibrosis patients (R-5), shear-rate-dependent modeling of 'mucociliary flow in mucus hypersecretion' (R-15), *Pseudomonas aeruginosa* effects on 'artificial mucus' transportability (R-28), electro-rheological ciliary flow of seminal fluid (R-24), and a mucus-inspired drug-loaded hydrogel (R-3). Despite recurring across five of the thirty top papers, none of the seven cluster labels mention 'mucus,' 'mucociliary,' or 'airway'; the theme is submerged inside the generic 'Rheology of Biological Fluids and Gels' label rather than recognized as its own thematic cluster.

Why: This is a framing gap: the keyword is frequent in titles/abstracts across five records but is not surfaced in any of the seven cluster labels.

Anchored in corpus papers:

- *Thixo-viscoelastoplastic rheological characterisation of human mucus using a multimode BMP constitutive equation approach* (2025) — Advanced mucus constitutive modeling with no dedicated cluster label
- **Pseudomonas aeruginosa* increases viscoelasticity and decreases transportability of artificial mucus* (2025) — Disease-linked mucus rheology unrepresented in cluster taxonomy

Candidate research questions:

- How has constitutive modeling of human mucus rheology (e.g., thixo-viscoelastoplastic BMP+ models) evolved to represent disease states such as Cystic Fibrosis? — *Consolidates scattered mucus-modeling studies into a coherent narrative of model development.*
Specific to this corpus: Targets mucus-focused records (R-5, R-15, R-28) currently absorbed into the 'Rheology of Biological Fluids and Gels' cluster label.
Suggested review type: narrative
- What is the relationship between bacterial infection (e.g., *Pseudomonas aeruginosa*) and measured viscoelastic/transport properties of airway mucus across published studies? — *Synthesizes infection-rheology linkage evident in the corpus.*
Specific to this corpus: Directly extends the finding in R-28 within the broader biological-fluids cluster.
Suggested review type: systematic

G-3: Constitutive Model Transfer Gap Mucus To Coronary CFD

Advanced thixotropic and viscoelastoplastic constitutive frameworks, such as the BMP+ model applied to mucus in R-5, have been developed for complex biofluids, yet the coronary/vascular CFD papers in the corpus - the IVUS-based stented coronary artery WSS study (R-7), the Cross-Williamson post-stenotic dilation model (R-22), and the pulsatile-tinnitus venous non-Newtonian comparison (R-23) - rely on simpler shear-thinning or power-law-type viscosity models rather than viscoelastoplastic/thixotropic formulations. This suggests the modeling sophistication present in the 'Rheology of Biological Fluids and Gels' cluster has not migrated into the 'Computational Hemodynamics In Coronary Arteries' cluster.

Why: *This is a method-application gap: a constitutive-modeling method (thixotropy/viscoelastoplasticity) is dominant in one cluster's biofluid work but is absent from the coronary/vascular CFD cluster where blood's known thixotropic behavior would plausibly matter.*

Anchored in corpus papers:

- *Thixo-viscoelastoplastic rheological characterisation of human mucus using a multimode BMP constitutive equation approach* (2025) — Source of advanced constitutive method not seen in coronary CFD papers
- *Influence of boundary conditions and blood rheology on indices of wall shear stress from IVUS-based patient-specific stented coronary artery* (2025) — Coronary CFD paper using simpler viscosity assumptions

Candidate research questions:

- How do thixotropic and viscoelastoplastic blood constitutive models compare with standard generalized-Newtonian models (e.g., Cross-Williamson, Carreau-Yasuda) in predicting wall shear stress in patient-specific coronary CFD? — *Directly tests whether more sophisticated rheological models change clinically relevant CFD outputs.*
Specific to this corpus: Targets the 'Computational Hemodynamics In Coronary Arteries' cluster (R-7, R-22) against modeling approaches from the biological-fluids cluster (R-5).
Suggested review type: systematic

- Has viscoelastic, as opposed to purely viscous, blood rheology been incorporated into patient-specific coronary or venous CFD simulations? — *Surveys the extent of model sophistication actually used in published vascular CFD work.*

Specific to this corpus: Covers R-7, R-22, R-23 within the coronary/venous computational hemodynamics clusters.

Suggested review type: scoping

G-4: Sparse Frontier Measurement Techniques 2025

Several 2025 records introduce novel low-frequency or contactless viscoelasticity measurement platforms - tensile acoustic rheometry (R-21), photothermal AFM nanorheology spanning '0.1-5000 Hz' (R-27), microrheology-on-a-chip using multiple particle tracking (R-13), and ultrasound-probe-pressure vessel-shape estimation of radial artery viscoelasticity (R-16) - each appearing as a single record among the top-30 with zero recorded citations. This indicates a burst of methodological innovation concentrated in the most recent year but insufficient replication within the corpus to determine a preferred technique for biological soft-tissue viscoelastic characterization.

Why: *This is a frontier gap: each technique bursts in the latest period yet has fewer than 10 associated records and no citation uptake, signaling an unconsolidated methodological frontier.*

Anchored in corpus papers:

- *Tensile acoustic rheometry for rapid and contactless characterization of soft viscoelastic biomaterials* (2025) — Single-record novel contactless technique with zero citations
- *Viscoelastic time responses of polymeric cell substrates measured continuously from 0.1-5000 Hz in liquid by photothermal AFM nanorheology* (2025) — Isolated new AFM-based method not cross-validated elsewhere

Candidate research questions:

- What is the comparative accuracy and reproducibility of emerging contactless viscoelastic measurement techniques (acoustic rheometry, photothermal AFM) for soft biological materials? — *Establishes a baseline evaluation before the frontier techniques diffuse further.*

Specific to this corpus: Targets the isolated 2025 method records R-21 and R-27 with no citation history yet.

Suggested review type: scoping

- How do novel in vivo/ex vivo viscoelasticity measurement platforms (ultrasound vessel-shape estimation, microfluidic microrheology) validate against established rheometric standards? — *Addresses whether new low-record-count techniques meet measurement validity criteria.*

Specific to this corpus: Covers R-16 and R-13 as emergent, sparsely replicated 2025 measurement approaches.

Suggested review type: systematic

G-5: Clinical Exercise Hemorheology Versus Computational Hemodynamics Disconnect

Clinical/physiological studies measuring blood rheological changes with exertion - 'Effects of regular exercise on blood rheology' comparing joggers versus sedentary controls (R-6), and 'Changes in the rheological properties of blood in combat sports athletes (boxing vs MMA)' (R-9) - report empirical hemorheological shifts but do not link these measured parameter changes to computational hemodynamic simulations. Conversely, the CFD-focused papers (R-7, R-22, R-23) use generic or assumed

non-Newtonian viscosity models rather than empirically measured, activity-dependent blood rheology parameters. This represents a bridge gap between the clinical hemorheology measurement literature and the computational hemodynamics literature within the corpus.

Why: *This is a bridge gap: two evidently related topics - measured blood rheology under physiological stress and simulated blood rheology in vascular CFD - rarely co-occur in the same records.*

Anchored in corpus papers:

- *Effects of regular exercise on blood rheology (2025)* — Empirical hemorheology data not fed into CFD models
- *Influence of boundary conditions and blood rheology on indices of wall shear stress from IVUS-based patient-specific stented coronary artery (2025)* — CFD study using assumed rather than measured rheology inputs

Candidate research questions:

- Do exercise-induced or activity-dependent changes in measured blood rheological parameters get incorporated as inputs into patient-specific computational hemodynamic models? — *Determines whether clinical measurement findings inform simulation parameterization.*

Specific to this corpus: Connects R-6/R-9 clinical hemorheology records to the coronary CFD cluster of R-7, R-22, R-23.

Suggested review type: systematic

- What is the range of hemorheological parameter values (viscosity, deformability) reported in exercise/sports physiology studies versus the fixed parameter values assumed in coronary CFD studies? — *Quantifies the magnitude of the disconnect between measured and assumed inputs.*

Specific to this corpus: Compares the 'Human Blood Rheology and Hemorheology' cluster with the 'Computational Hemodynamics In Coronary Arteries' cluster.

Suggested review type: scoping

G-6: Synovial Fluid Viscosupplementation Isolated From Modeling Clusters

Clinical intervention studies on synovial fluid rheology in knee osteoarthritis - the polynucleotide/BDDE-crosslinked hyaluronic acid characterization for 'intra-articular viscosupplementation' (R-4) and 'Enhanced Synovial Fluid Rheology in Moderate Knee Osteoarthritis Through Combined Intra-Articular Hyaluronic Acid and Multimodal Physiotherapy' (R-30) - focus on clinical/rheometric characterization of treatment outcomes but show no linkage to the constitutive-modeling or computational simulation approaches used elsewhere in the corpus, such as BMP+ mucus modeling (R-5) or coronary CFD (R-7, R-22). This isolates synovial fluid viscosupplementation as a clinically-oriented offshoot with no apparent methodological cross-pollination from the rheological-modeling clusters.

Why: *This is a method-application gap: constitutive/computational modeling methods well established in other clusters of this corpus are absent from synovial fluid treatment-outcome studies where they could quantify viscosupplementation efficacy.*

Anchored in corpus papers:

- *Comprehensive rheological characterization of polynucleotide/BDDE-crosslinked hyaluronic acid blend for intra-articular viscosupplementation* (2025) — Clinical rheometric study with no constitutive/computational modeling link
- *Enhanced Synovial Fluid Rheology in Moderate Knee Osteoarthritis Through Combined Intra-Articular Hyaluronic Acid and Multimodal Physiotherapy* (2025) — Treatment-outcome rheology isolated from modeling clusters

Candidate research questions:

- Could constitutive viscoelastic models developed for other biological gels/fluids be applied to quantify hyaluronic-acid-based viscosupplementation efficacy in knee osteoarthritis? — *Explores cross-application of modeling frameworks already validated elsewhere in the corpus.*
Specific to this corpus: Links R-4/R-30 synovial fluid studies to constitutive-modeling approaches like R-5.
Suggested review type: narrative
- What rheological metrics are used across intra-articular hyaluronic acid studies to define 'restored' synovial fluid viscoelasticity, and how do they compare to native synovial fluid benchmarks? — *Synthesizes outcome-measurement standards that are currently fragmented across isolated clinical studies.*
Specific to this corpus: Focuses on the synovial fluid viscosupplementation records (R-4, R-30) within the biological-fluids cluster.
Suggested review type: systematic

1.3 Corpus coverage notes

Strengths: Rheological and viscosity-focused characterization of biological fluids is a core strength: this theme dominates the top-keyword list (rheology, viscosity, non-Newtonian fluids) across the full 2016–2025 span of the 697-record corpus and 388 distinct publication sources.; Hemodynamics and blood-flow viscoelasticity form a second deeply anchored theme, co-occurring with 'blood flow', 'hemodynamics', and 'viscoelastic properties' among the top 15 keywords spanning the same decade (2016–2025) and broad venue base of 388 sources.; Human and animal model-based viscoelasticity studies (keywords 'humans', 'animals', 'models', 'elasticity') recur consistently enough to rank in the top 15 terms across all 697 records and the full 2016–2025 timeframe, indicating durable methodological grounding.

2. Publication trends

The corpus contains 697 records spanning 2016 to 2025, drawn from 2634 unique authors and 3888 unique keywords. Over the observed window, annual output grew at a compound rate of 6.4%, a steady expansion of the field. Change point analysis identified 2 distinct publication regimes: 2016–2020 averaged 51.6 publications per year; 2021–2025 averaged 87.8 publications per year. The leading contributing countries are United States (158), India (93), China (72). Contributions are geographically distributed, with no single country dominating the literature. Publications appear across 25 distinct venues, led by Physics of Fluids with 28 records.

Publication output across the 697-record corpus rose from 52 in 2016 to 101 in 2024, with the trajectory punctuated by a dip in 2019 (44 records, down from 54 in 2018) followed by a sharp rebound and acceleration into 2021 (76) and a peak in 2022 (94), before a slight pullback to 77 in 2023 and a strong 101 in 2024 and 91 in 2025. Era-level averages confirm this shift: the 2016–2020 era averaged 51.6 publications/year, while the 2021–2025 era averaged 87.8 publications/year, a marked step-change in output. The corpus-wide CAGR of 6.4% over the 2016–2025 span indicates steady but moderate compounding growth rather than explosive expansion; at this rate the field's annual output would take roughly 11 years to double. Geographically, the United States leads with 158 records (20.5%), followed by India with 93 (12.1%) and China with 72 (9.4%); together the top-3 countries account for 42.0% of records, suggesting moderate geographic concentration with substantial contribution from a long tail of other nations (United Kingdom, Italy, Pakistan, France, Germany, Iran, Saudi Arabia each between 3.9% and 7.4%). Venue-wise, Physics of Fluids leads with 28 records, followed by Scientific Reports with 16, indicating a dispersed publishing landscape rather than dominance by one or two outlets.

Figure 1 shows annual publication counts drawn directly from the 697 records in the provided corpus. Each bar represents the number of papers published in that year. Coloured horizontal lines mark the mean publication rate for each algorithmically detected research era; shaded background regions delineate the era boundaries identified by change point analysis.

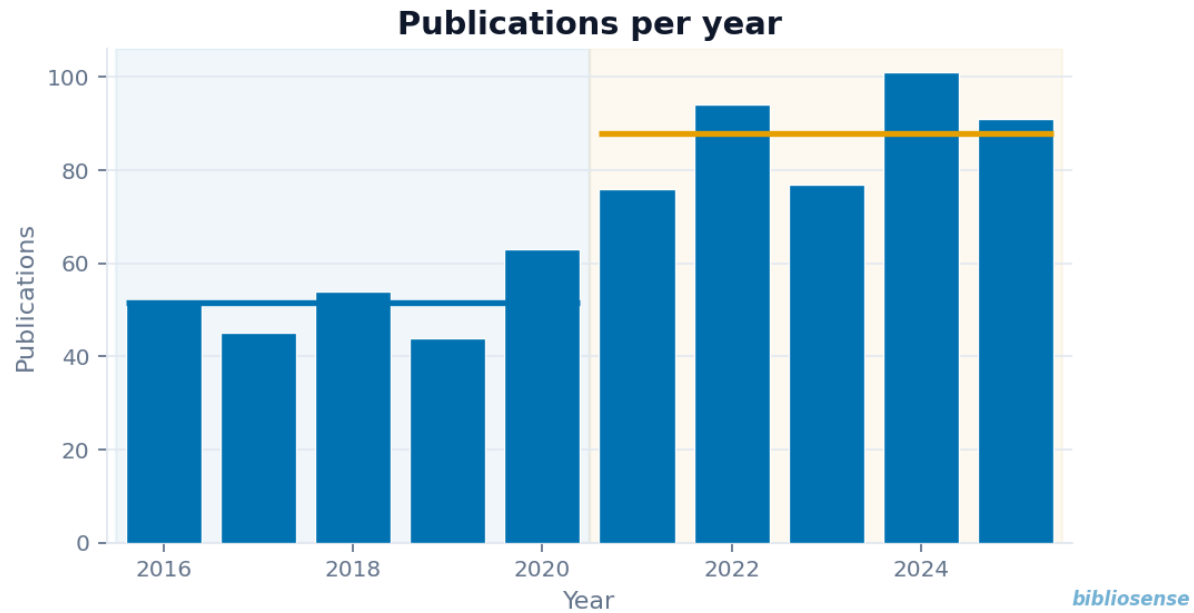


Figure 1. Annual publications per year (bars) with changepoint-detected research eras. Horizontal lines = mean rate per era; shaded regions = era spans.

Figure 2 shows the top contributing countries by number of records in the provided corpus, derived from author affiliations.

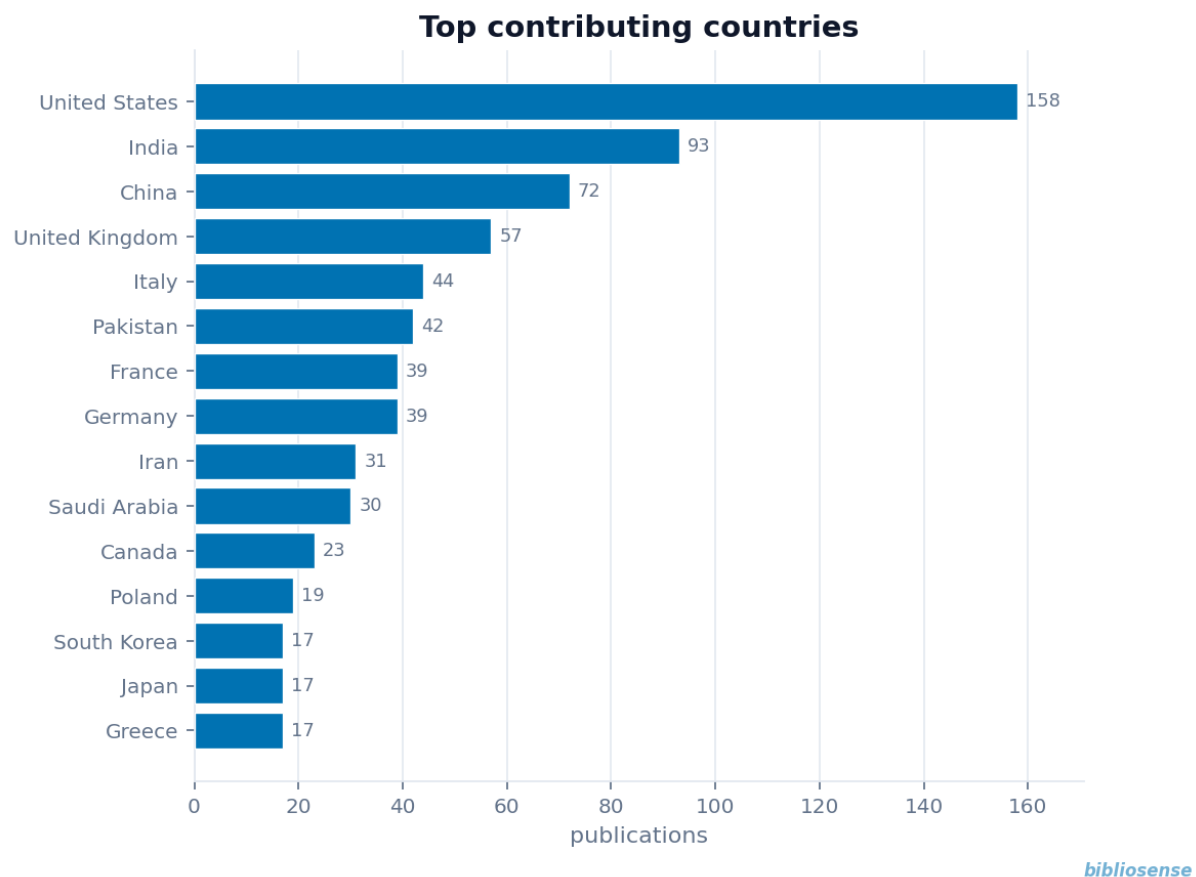


Figure 2. Leading contributing countries by publication count.

Of 696 records with parsed affiliations, 29% involve international collaboration (at least two distinct countries among the author affiliations). United States leads output; 43% of its records are internationally co-authored, most frequently with China (13 joint records), United Kingdom (9 joint records), India (8 joint records), Pakistan (7 joint records). The strongest country-country collaboration ties are: Pakistan–Saudi Arabia (19); China–United States (13); United Kingdom–United States (9).

Table 1. Top publication venues

venue	publications
Physics of Fluids	28
Scientific Reports	16
Clinical Hemorheology and Microcirculation	15
Soft Matter	13
Journal of Fluid Mechanics	10
Computer Methods and Programs in Biomedicine	10
Frontiers in Physiology	9
Journal of Non Newtonian Fluid Mechanics	9
Physical Review Fluids	8
Rheologica Acta	7
International Journal for Numerical Methods in Biomedical Engineering	7
Journal of the Mechanical Behavior of Biomedical Materials	7
Journal of Computational Physics	6
Fluids	6
Heat Transfer	6

Adaptive thresholds used

View	Parameter	Value	Rationale
venues	min_count	2.00	Corpus n=697 → heuristic strategy. Stripped 1 dominant head item (corpus-topic terms) before knee detection. Kneedle knee at rank

			387 of 387 (after stripping 1 dominant head item); items with count ≥ 2 survive (109 of 388).
countries	min_count	7.00	Corpus n=697 $\rightarrow$ heuristic strategy. Kneedle knee at rank 1 of 77 (after stripping 1 dominant head item) would have left only 2 items; floored to top-30 cutoff (count ≥ 7 , 30 surviving).

3. Keyword co-occurrence network

The keyword co-occurrence network, after disparity-filter backbone extraction, comprises 99 references and 258 edges. Leiden modularity optimisation identified 7 distinct clusters. The most central references are: humans; rheology; viscosity; models; stress. Principal clusters include cluster 1 (33 references) anchored by rheology and viscosity; cluster 2 (22 references) anchored by blood flow and non-newtonian fluids; cluster 3 (21 references) anchored by humans and male.

The keyword map groups the corpus into 7 colour-coded communities — the colours in the legend. Larger nodes are the more frequent terms, the ringed nodes are the hubs that most other terms connect through, and links mark terms that co-occur; the denser and more saturated a region, the more it anchors the field. It is the same structure the interactive explorer shows, fixed here for the record. The terms that hold the whole map together are “humans”, “rheology”, “viscosity” — the shared vocabulary the surrounding clusters all reach into. Reading the communities in turn: Computational Hemodynamics In Coronary Arteries (led by “models”, “stress”, “mechanical”); Rheology of Biological Fluids and Gels (led by “rheology”, “viscosity”, “animals”); Human Blood Rheology and Hemorheology (led by “humans”, “male”, “female”); Non-Newtonian Blood Flow Haemodynamics (led by “blood flow”, “non-newtonian fluids”, “blood rheology”) — these are the centres of gravity to orient around before drilling into individual papers.

The network contains 99 unique keywords linked by 258 co-occurrence edges, an edge-to-node ratio near 2.6 that signals moderately tight integration: most terms cluster into recurring pairings rather than standing alone, yet the corpus still resolves into 7 distinct communities, indicating a literature unified by a shared rheological vocabulary but split across several methodological sub-themes. Each cluster is identifiable in the records. “Rheology of Biological Fluids and Gels” (33 keywords), the largest, is exemplified by “Red cells' dynamic morphologies govern blood shear thinning under microcirculatory flow conditions” (2016), which reports RBC tumbling and rolling behaviors governing shear thinning. “Non-Newtonian Blood Flow Haemodynamics” (22 keywords) is represented by “A review on non-Newtonian fluid models for multi-layered blood rheology in constricted arteries” (2023). “Human Blood Rheology and Hemorheology” (21 keywords) includes “Simulation of Blood as Fluid: A Review from Rheological Aspects” (2021). “Computational Hemodynamics In Coronary Arteries” (16 keywords) contains “Investigating Blood's Shear-Thinning Effects While Flowing Through a Post-Stenotic Dilated Artery: A Cross–Williamson Model Approach” (2025), which develops governing equations via CFD/FEM for stenosed arteries. The smaller “Viscoelastic Fluid Flow Dynamics” (3 keywords) is shown by “Nonwetting droplet oscillation and displacement by viscoelastic fluids” (2020). “Computational Hemodynamics and Non-Newtonian Flow” (2 keywords) is captured by the review “Image-Based Hemodynamic and Rheological Study of Patient's Diseased Arterial Vasculatures...” (2025). “Rheological Modeling of Biological Fluids” (2 keywords) is exemplified by “How viscoelastic is human blood plasma?” (2018), which computes plasma's “viscoelastic properties in shear and extensional flows.” The top hub keywords, humans (846), rheology (665), and viscosity (557), reveal that this corpus centers on human-relevant characterization of fluid behavior, as seen in “Effects of regular exercise on blood rheology” (2025), which measured “blood viscosity, plasma viscosity, and red blood cell aggregation” in human participants. Among bridge terms, “models” links the Computational Hemodynamics In Coronary Arteries

mechanical	39	14	274	3
stress, mechanical	38	14	274	3
cardiovascular	32	12	231	3
models, cardiovascular	32	12	231	3
hemodynamics	64	15	230	3
male	50	11	212	2
animals	60	15	207	0
elasticity	53	8	162	0
blood flow velocity	29	10	159	3
female	32	7	132	2

Adaptive thresholds used

View	Parameter	Value	Rationale
keyword_cooccurrence	min_count	5.00	Corpus n=697 → heuristic strategy. Kneedle knee at rank 209 of 209 (after stripping 0 dominant head items); items with count ≥ 5 survive (209 of 209).
keyword_cooccurrence	edge_backbone_alpha	0.05	Serrano et al. disparity filter at $\alpha=0.05$: kept 258/4441 edges.

4. Collaboration networks

4.1 Author collaboration

The author collaboration network, after disparity-filter backbone extraction, comprises 47 references and 24 edges. Leiden modularity optimisation identified 23 distinct clusters. The network is sparse (edge density 0.022), suggesting distinct specialisations with limited cross-connection. The most central references are: Mahapatra, B; Shah, Z; Bandopadhyay, A; Krztoń-Maziopa, A; Mrokowska, MM. Principal clusters include cluster 1 (3 references) anchored by Shah, Z and Shutaywi, M; cluster 2 (2 references) anchored by Shaw, S and Tripathi, D; cluster 3 (2 references) anchored by Schmidt, TA and Martin-Alarcon, L.

Figure 4 shows the author collaboration backbone; each node is an author and each edge represents a co-authored record.

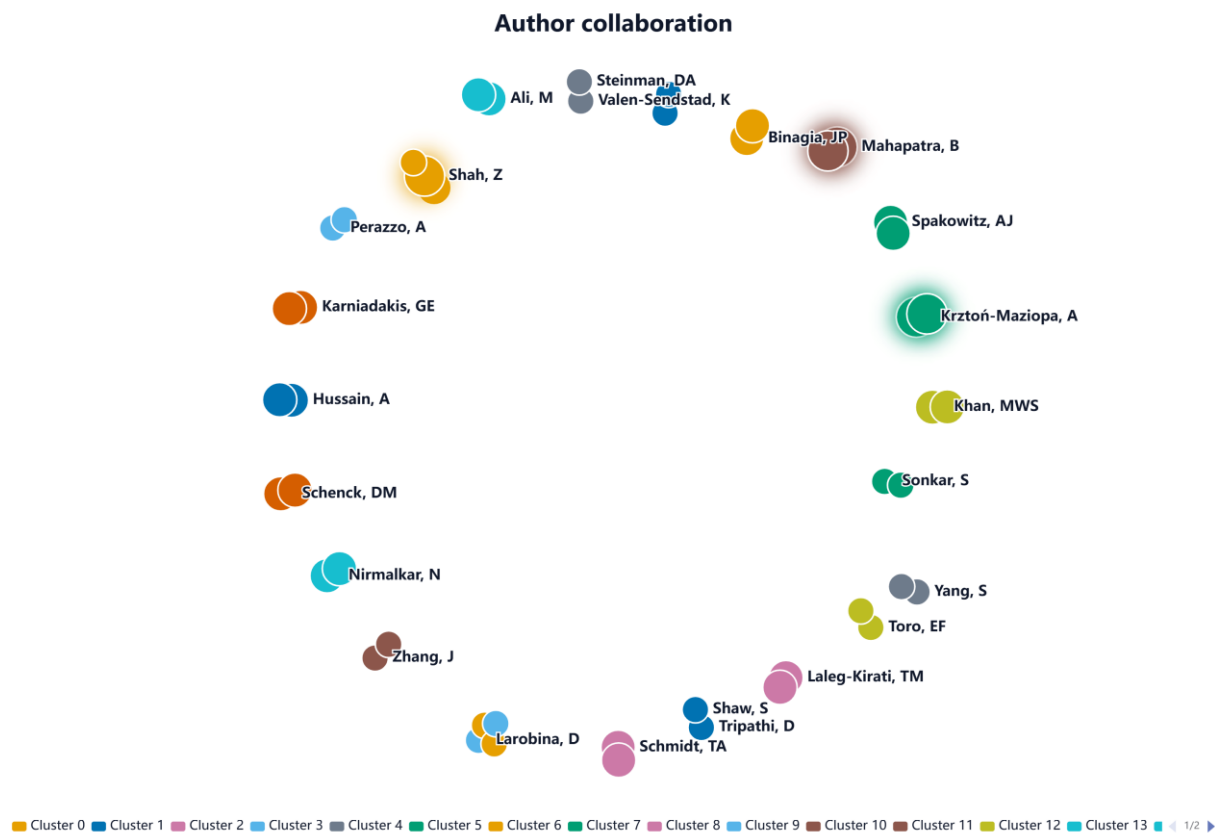


Figure 4. Author collaboration network backbone.

Adaptive thresholds used

View	Parameter	Value	Rationale
author_collaboration	edge_backbone_alpha	0.05	Serrano et al. disparity filter at $\alpha=0.05$: kept 24/776 edges.

4.2 Institution collaboration

The institution collaboration network, after disparity-filter backbone extraction, comprises 104 references and 55 edges. Leiden modularity optimisation identified 49 distinct clusters. The network is sparse (edge density 0.010), suggesting distinct specialisations with limited cross-connection. The most central references are: Universidade do Porto; UNC School of Medicine; Chapel Hill; Instituto de Ciência e Inovação em Engenharia Mecânica e Engenharia Industrial; Paris. Principal clusters include cluster 1 (3 references) anchored by Foundation for Research and Technology-Hellas and Politecnico di Milano; cluster 2 (3 references) anchored by Indian Institute of Technology Kharagpur and Indian Institute of Technology Guwahati; cluster 3 (3 references) anchored by A. James Clark School of Engineering and The University of Tokyo.

Figure 5 shows the institution collaboration backbone. Nodes represent institutions; edges represent shared authorship across records.

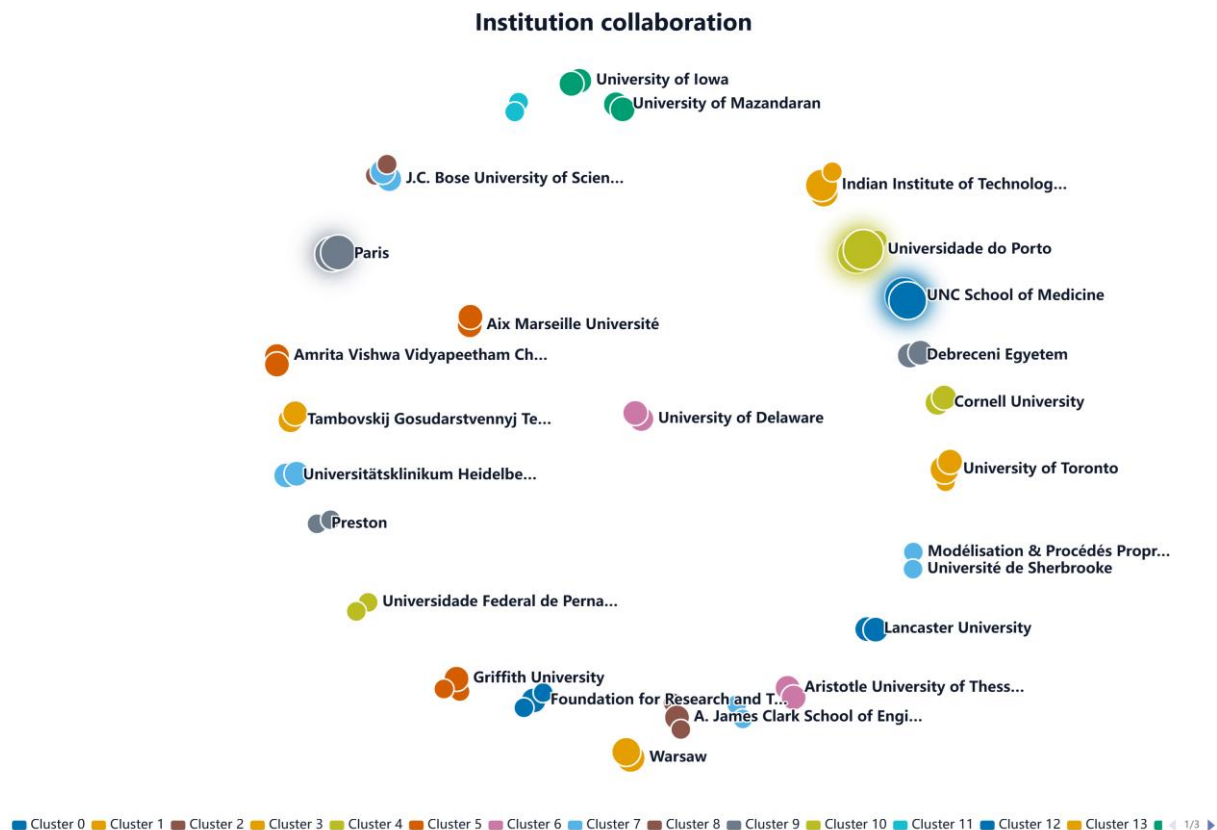


Figure 5. Institution collaboration network backbone.

Adaptive thresholds used

View	Parameter	Value	Rationale
institution_collaboration	edge_backbone_alpha	0.05	Serrano et al. disparity filter at $\alpha=0.05$: kept 55/462 edges.

4.3 Country / regional collaboration

Affiliations aggregated to country level. This view is denser than the institution-level one because every paper with authors in two or more countries contributes a country-pair edge — useful for revealing geographic research blocs that institution-level analysis can hide when most papers are intra-institutional.

The country collaboration network, after disparity-filter backbone extraction, comprises 33 references and 22 edges. Leiden modularity optimisation identified 12 distinct clusters. The network is sparse (edge density 0.042), suggesting distinct specialisations with limited cross-connection. The most central references are: United States; United Kingdom; Pakistan; Saudi Arabia; India. Principal clusters include cluster 1 (9 references) anchored by United States and United Kingdom; cluster 2 (3 references) anchored by Pakistan and Saudi Arabia; cluster 3 (3 references) anchored by France and New Zealand.

Figure 6 maps the country-level collaboration network. Edge thickness reflects the number of jointly-authored records.

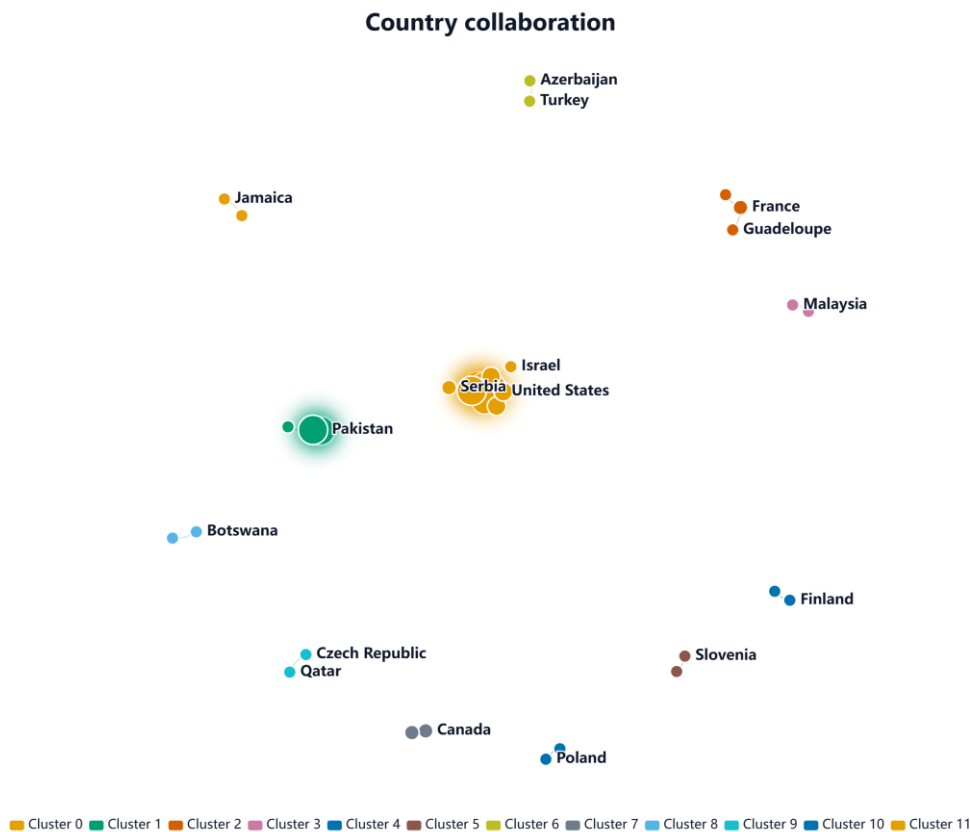


Figure 6. Country / regional collaboration network.

Adaptive thresholds used

View	Parameter	Value	Rationale
country_collaboration	edge_backbone_alpha	0.05	Serrano et al. disparity filter at $\alpha=0.05$: kept 22/214 edges.

5. Citation-based networks

5.0 Seminal references

The following references are cited most frequently across the corpus. Of 25258 distinct cited references, 15 are cited by five or more records, marking them as the intellectual backbone of this literature. Each reference is tagged with the co-citation cluster it anchors, indicating which sub-community of the field it belongs to.

Note: the count in the “Citing records (in corpus)” column is the number of papers in this 697-record corpus that cite each reference — not the absolute citation count from Scopus. For an SLR this is the meaningful quantity: it identifies which earlier works the literature under review repeatedly anchors on. Absolute citation counts for corpus papers themselves appear in §8 (Citation impact).

Table 3 lists the 15 most-cited references within the corpus together with the co-citation cluster each anchors.

#	Citing records (in corpus)	Reference	Co-citation cluster anchored
1	54	SCOPUS:60149100339	Cluster 2: SCOPUS:60149100339 / SCOPUS:84895871605 / SCOPUS:84896741711
2	50	SCOPUS:0003423425	Cluster 4: SCOPUS:0003423425 / SCOPUS:84949213211 / SCOPUS:70350589080
3	45	SCOPUS:0025768839	Cluster 1: SCOPUS:0025768839 / SCOPUS:1842471932 / SCOPUS:79551494253
4	35	SCOPUS:1842471932	Cluster 1: SCOPUS:0025768839 / SCOPUS:1842471932 / SCOPUS:79551494253
5	33	SCOPUS:0242330201	Cluster 5: SCOPUS:84880114474 / SCOPUS:84874065619

			/ SCOPUS:0242330201
6	27	SCOPUS:0015402207	Cluster 3: SCOPUS:0015402207 / SCOPUS:79961050456 / SCOPUS:26444596543
7	25	SCOPUS:0003715001	Cluster 1: SCOPUS:0025768839 / SCOPUS:1842471932 / SCOPUS:79551494253
8	24	SCOPUS:84874065619	Cluster 5: SCOPUS:84880114474 / SCOPUS:84874065619 / SCOPUS:0242330201
9	23	SCOPUS:79551494253	Cluster 1: SCOPUS:0025768839 / SCOPUS:1842471932 / SCOPUS:79551494253
10	21	SCOPUS:0032953847	Cluster 1: SCOPUS:0025768839 / SCOPUS:1842471932 / SCOPUS:79551494253
11	21	SCOPUS:84880114474	Cluster 5: SCOPUS:84880114474 / SCOPUS:84874065619 / SCOPUS:0242330201
12	20	SCOPUS:0001331302	Cluster 3: SCOPUS:0015402207 / SCOPUS:79961050456 / SCOPUS:26444596543
13	20	SCOPUS:0001899137	Cluster 3: SCOPUS:0015402207 / SCOPUS:79961050456 / SCOPUS:26444596543

14	20	SCOPUS:34848909567	Cluster 1: SCOPUS:0025768839 / SCOPUS:1842471932 / SCOPUS:79551494253
15	19	SCOPUS:79960326068	Cluster 1: SCOPUS:0025768839 / SCOPUS:1842471932 / SCOPUS:79551494253

5.1 Co-citation

The co-citation network, after disparity-filter backbone extraction, comprises 385 references and 731 edges. Leiden modularity optimisation identified 25 distinct clusters. The network is sparse (edge density 0.010), suggesting distinct specialisations with limited cross-connection. The most central references are: SCOPUS:0025768839 [cited by 45 records]; SCOPUS:60149100339 [cited by 54 records]; SCOPUS:1842471932 [cited by 35 records]; SCOPUS:0003423425 [cited by 50 records]; SCOPUS:79551494253 [cited by 23 records]. Principal clusters include cluster 1 (75 references, theme: scopus) anchored by SCOPUS:0025768839 and SCOPUS:1842471932; cluster 2 (73 references, theme: scopus) anchored by SCOPUS:60149100339 and SCOPUS:84895871605; cluster 3 (42 references, theme: scopus) anchored by SCOPUS:0015402207 and SCOPUS:79961050456.

Figure 7 visualises the co-citation backbone. References that are frequently cited together cluster together; the colour indicates the Leiden community each reference belongs to.

Co-citation

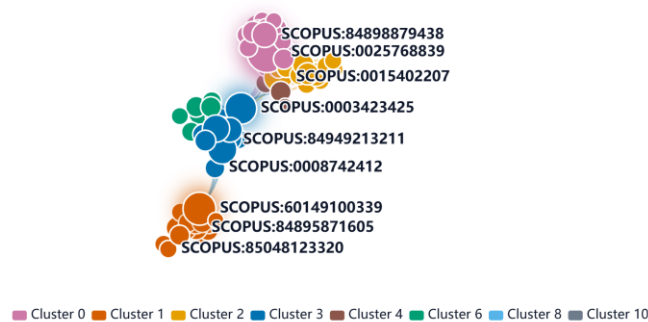


Figure 7. Co-citation network backbone.

Adaptive thresholds used

View	Parameter	Value	Rationale
cocitation	min_count	2.00	Corpus n=697 → heuristic strategy. Kneedle knee at rank 4093 of 4093 (after stripping 0 dominant head items); items with count ≥ 2 survive (4093 of 4093).
cocitation	edge_backbone_alpha	0.05	Serrano et al. disparity filter at $\alpha=0.05$: kept 731/164770 edges.

5.2 Bibliographic coupling

The bibliographic coupling network, after disparity-filter backbone extraction, comprises 179 papers and 96 edges. Leiden modularity optimisation identified 83 distinct clusters. The network is sparse (edge density 0.006): few papers in the corpus share enough cited references to form strong couplings, which typically reflects a topic-broad corpus where authors draw on substantially different bibliographies. The most central papers are: (2020) “Non-Linear or Quasi-Linear Viscoelastic Property of Blood for Hemodynamic Simulations”, Advanced Structured Materials; (2019) “Rheological analysis of blood flow in the bifurcation of carotid artery with openFOAM”, Smart Innovation Systems and Technologies; (2021)

“Computational Fluid Dynamics Analysis of Blood Rheology”, Smart Innovation Systems and Technologies; (2020) “WSS Descriptors in a Patient RCA Taking into Account the Non-linear Viscoelasticity of ...”, Advanced Structured Materials; (2024) “Viscoelastic characterization of the mucus from the skin of loach (II) using linear vis...”, Journal of the Chinese Institute of Engineers Transactions of the Chinese Institute of Engineers Series A. Principal clusters include cluster 1 (4 papers, theme: blood/rheology) anchored by (2019) “Blood rheology: Key parameters, impact on blood flow, role in sickle cell disease and e...” and (2020) “Blood rheology biomarkers in sickle cell disease”; cluster 2 (4 papers, theme: properties/synovial/effect) anchored by (2024) “THE EFFECT OF THE MOLECULAR WEIGHT OF HYALURONIC ACID ON THE RHEOLOGICAL AND TRIBOLOGIC...” and (2021) “Rheological properties and its effect on the lubrication mechanism of pvp k30 and pvp 4...”; cluster 3 (3 papers, theme: scopus/characterization/dynamic) anchored by (2017) “Linear and Nonlinear Viscoelastic Arterial Wall Models: Application on Animals” and (2021) “Characterization of the dynamic viscoelastic response of the ascending aorta imposed vi...”.

Figure 8 shows the bibliographic coupling network. Papers that share many cited references are linked; tightly coupled papers typically address closely related sub-problems.

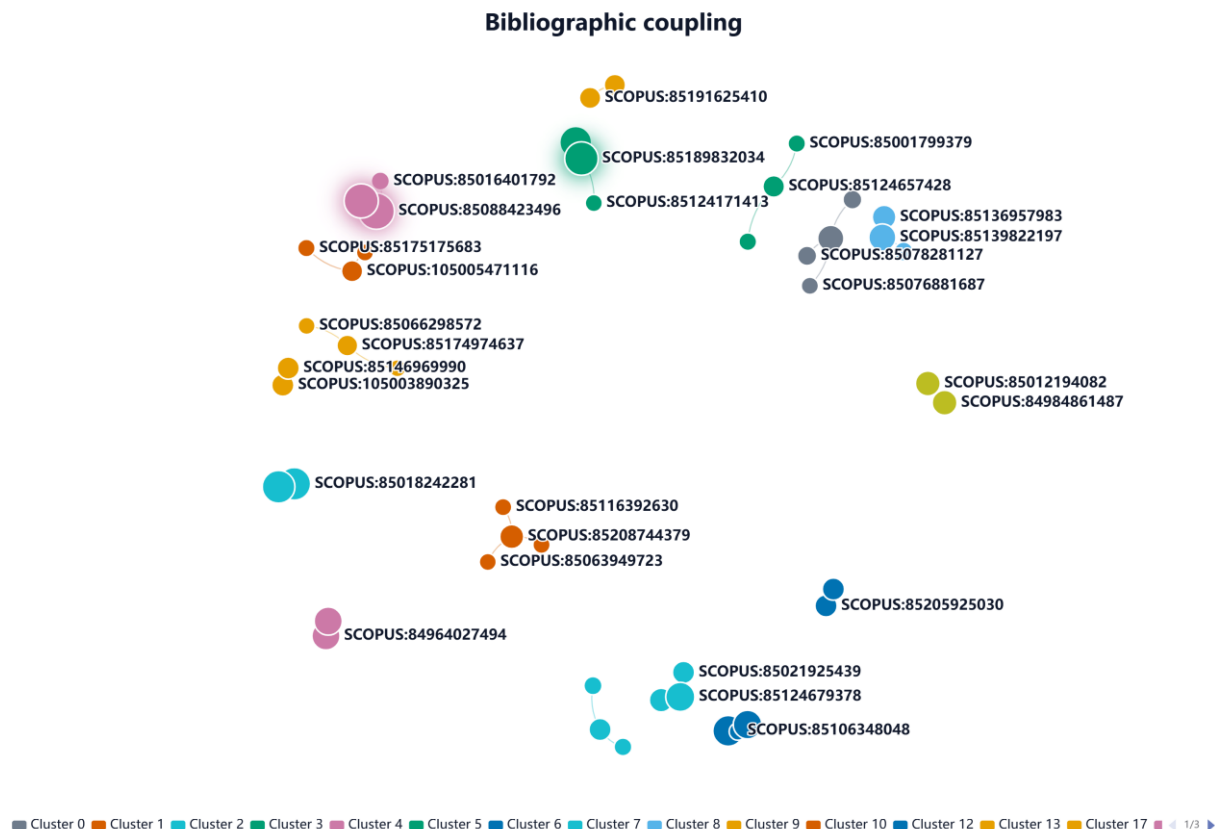


Figure 8. Bibliographic coupling network.

Adaptive thresholds used

View	Parameter	Value	Rationale
bibliographic_coupling	edge_backbone_alpha	0.05	Serrano et al. disparity filter at $\alpha=0.05$:

			kept 96/1231 edges.
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6. Thematic evolution

The corpus traces four eras of research on non-Newtonian blood rheology and computational hemodynamics. In 2016–2018, the field was anchored by rheology (10.3% share), humans (8.8%), and viscosity (8.1%), reflecting a foundational focus on characterizing biological fluids at the bench and in early computational models. Landmark works from this period include Lanotte et al.'s "Red cells' dynamic morphologies govern blood shear thinning under microcirculatory flow conditions" (213 citations), which linked RBC dynamics to shear thinning, and Arzani's "Accounting for residence-time in blood rheology models," which questioned whether non-Newtonian modelling was necessary in large arteries. Mucus and synovial-fluid rheology also featured prominently, as in Wagner et al.'s study of MUC5AC gels and Nicholls et al.'s comparison of hyaluronic acid preparations against synovial fluid.

Moving into 2019–2021, rheology's share rose slightly (10.3%→10.6%) while humans held steady (8.8%→8.8%) and viscosity declined (8.1%→7.4%). This era is exemplified by Nader et al.'s highly cited "Blood rheology: Key parameters, impact on blood flow, role in sickle cell disease and effects of exercise" (429 citations) and Kannojiya et al.'s "Simulation of Blood as Fluid: A Review from Rheological Aspects," both consolidating rheological knowledge into synthesis and simulation frameworks, alongside growing CFD applications such as Mendieta et al.'s patient-specific carotid stenosis simulations.

The 2022–2024 period marks the most dramatic shift: viscoelastics surged from 0.8% to 8.1% share, and blood flow rose from 4.2% to 7.5%, while rheology edged down (10.6%→10.0%). This viscoelastic turn is visible in Gonzalo et al.'s "Non-Newtonian blood rheology impacts left atrial stasis in patient-specific simulations" and De Nisco et al.'s "Modelling blood flow in coronary arteries: Newtonian or shear-thinning non-Newtonian rheology?", which directly interrogate rheological assumptions in hemodynamic models, alongside Schaupp et al.'s clinical study of sputum viscoelastic properties in cystic fibrosis (163 citations).

By 2025, rheology (8.8%), humans (8.3%), and viscosity (6.9%) remain the top descriptors but all declined from 2022–2024 levels, while viscoelastics fell sharply (8.1%→4.4%) and blood flow eased (7.5%→5.4%), even as demographic and mechanical descriptors like elasticity and male gained ground—visible in patient-specific studies such as McCarthy et al.'s IVUS-based stented coronary artery simulations.

Strategic diagram (thematic map)

Figure 9 positions each keyword cluster on Callon's strategic diagram. The horizontal axis is centrality — how strongly a theme links to the rest of the field; the vertical axis is density — how internally developed the theme is. Upper-right clusters are MOTOR themes (central and mature); upper-left are NICHE themes (specialised but peripheral); lower-right are BASIC or transversal themes the field is built on; lower-left are EMERGING or declining themes.

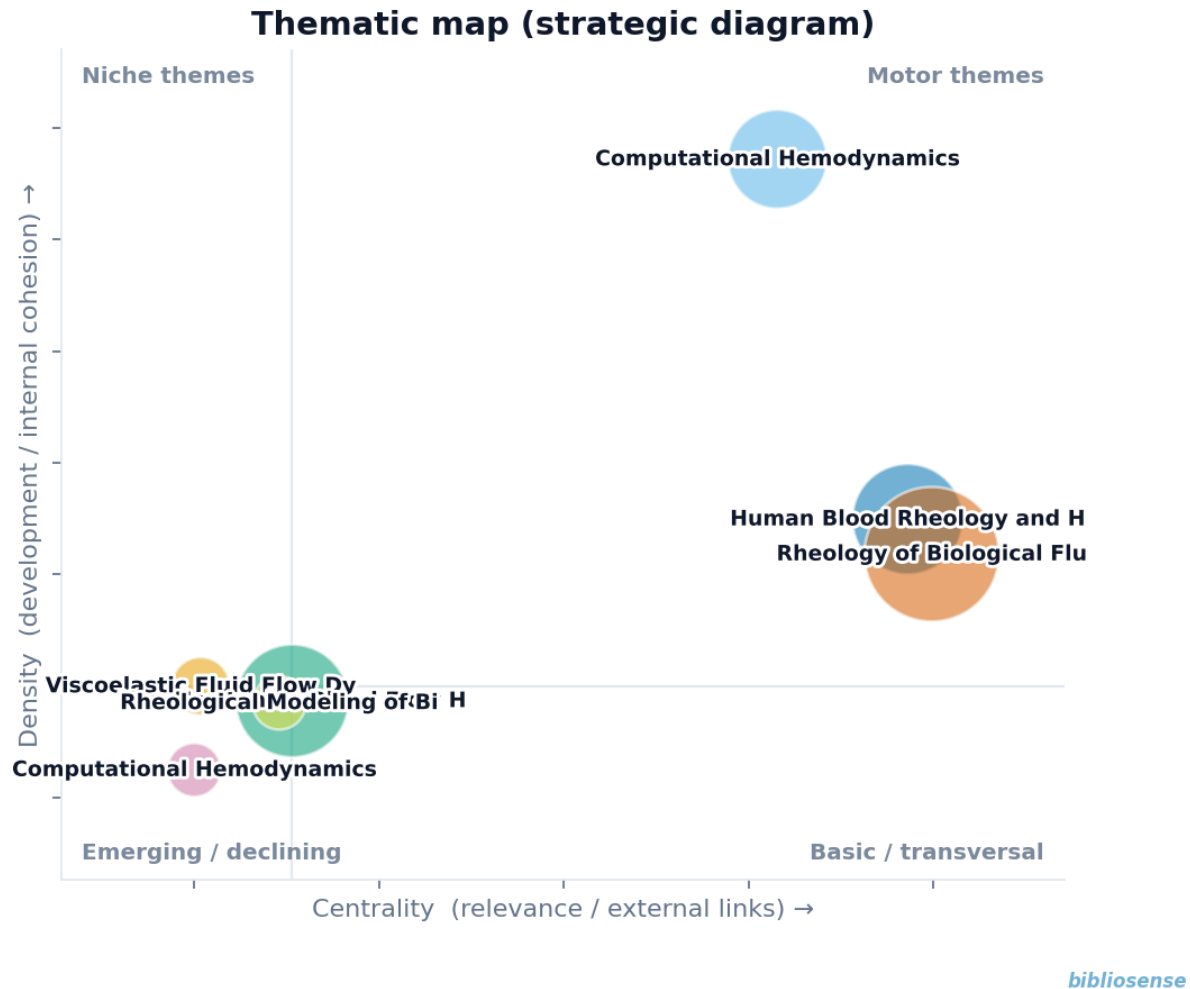


Figure 9. Thematic map (strategic diagram) of the keyword clusters.

Motor themes: Human Blood Rheology and Hemorheology; Rheology of Biological Fluids and Gels; Computational Hemodynamics In Coronary Arteries.

Basic / transversal themes: Non-Newtonian Blood Flow Haemodynamics.

Niche themes: Viscoelastic Fluid Flow Dynamics.

Emerging or declining themes: Computational Hemodynamics and Non-Newtonian Flow; Rheological Modeling of Biological Fluids.

Comparing 2022–2024 with 2025, the most pronounced shift is toward 'mechanical' (rising from 2.2% to 4.9% of period keyword share), alongside a contraction in 'viscoelastics' (from 12.2% to 7.4%).

Figure 10 is a keyword-intensity heatmap: rows are the top keywords retained above the adaptive cutoff, columns are the detected time periods, and cell shade encodes relative frequency within each period. Rows that darken from left to right signal emerging themes; rows that fade signal declining ones.

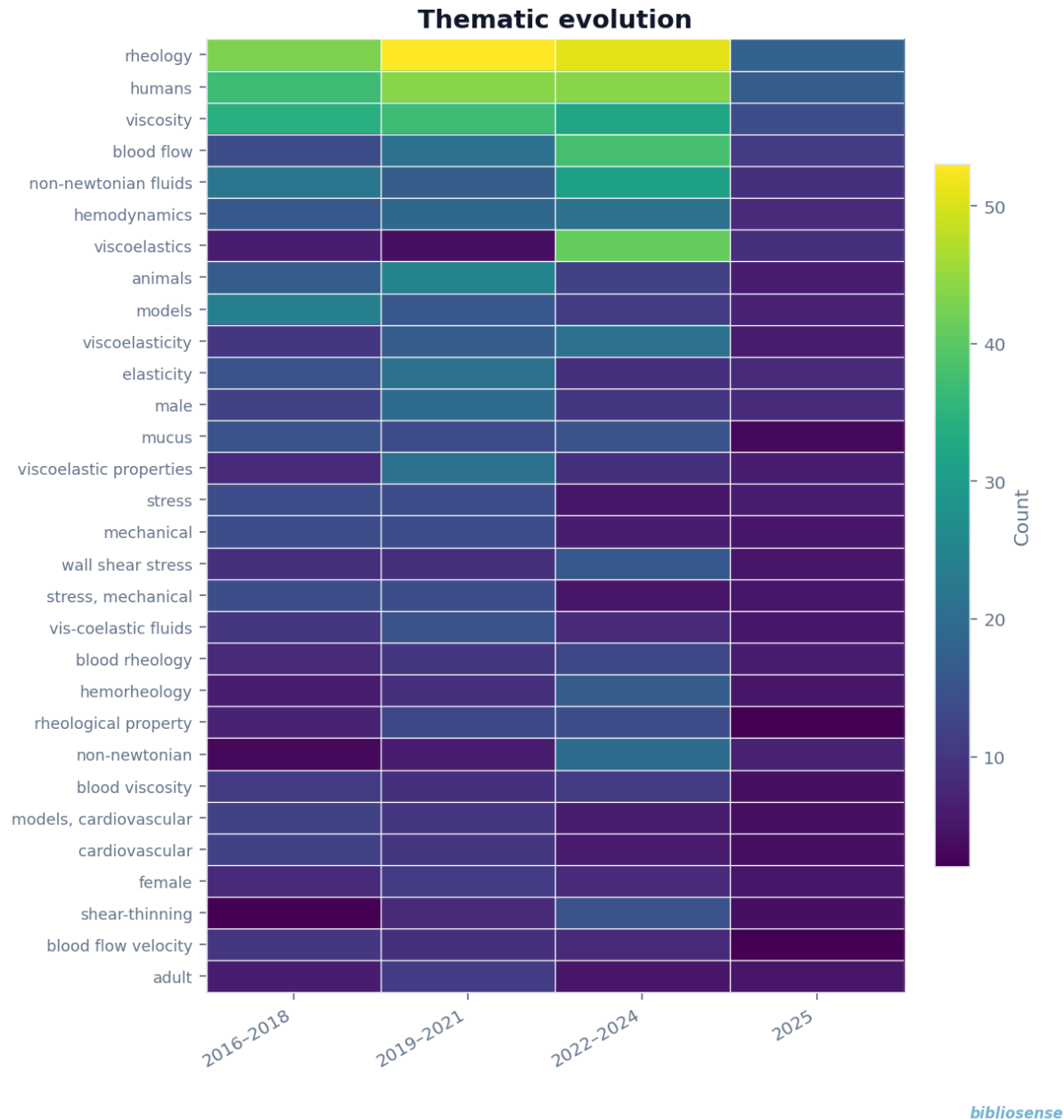


Figure 10. Keyword intensity by period (row = keyword, column = period, shade = relative frequency).

6.1 Transitions between the most recent periods

Table 4 compares keyword share between 2022–2024 and 2025. Topics are grouped via lightweight semantic clustering (abbreviations resolved, plural and hyphen variants merged) so that 'CNN' and 'convolutional neural network' count as a single topic. Persistent topics stayed strong in both periods; fading topics lost share; emerging topics rose sharply.

Comparing 2022–2024 (previous period) with 2025 (most recent).

Persistent topics: Strong in both periods; the enduring backbone of the field.

Topic	2022–2024 share	2025 share	Change
rheology	12.6%	11.8%	-0.8pp
viscosity	8.5%	8.8%	+0.4pp
humans	8.7%	8.3%	-0.3pp
male	2.0%	3.9%	+2.0pp
cardiovascular	2.4%	3.9%	+1.6pp
hemodynamics	4.1%	3.9%	-0.2pp
models	2.2%	3.4%	+1.3pp
animals	2.4%	2.9%	+0.6pp
viscoelastic properties	1.8%	2.9%	+1.2pp
hemorheology	3.3%	2.5%	-0.9pp

Fading topics: Lost share; interest may be waning.

Topic	2022–2024 share	2025 share	Change
viscoelastics	12.2%	7.4%	-4.9pp
non-newtonian	10.0%	7.8%	-2.2pp
blood flow	7.5%	5.4%	-2.1pp
rheological property	2.8%	1.0%	-1.8pp
mucus	3.0%	1.5%	-1.5pp

Emerging topics: Rose sharply in the most recent period; current frontier.

Topic	2022–2024 share	2025 share	Change
mechanical	2.2%	4.9%	+2.7pp
elasticity	1.8%	3.9%	+2.1pp
stress	1.0%	2.9%	+2.0pp
male	2.0%	3.9%	+2.0pp
cardiovascular	2.4%	3.9%	+1.6pp
adult	1.0%	2.5%	+1.5pp
models	2.2%	3.4%	+1.3pp
viscoelastic properties	1.8%	2.9%	+1.2pp

Between 2022–2024 and 2025, the transition table shows rheology declining from 10.0% to 8.8% share and viscoelastics falling sharply from 8.1% to 4.4%, alongside declines in blood flow (7.5%→5.4%), non-newtonian fluids (6.1%→4.4%), and mucus (3.0%→1.5%). Mucus's fading is illustrated by earlier dominant work such as Demouveau et al.'s 2018 "Gel-forming mucin interactome drives mucus viscoelasticity," when mucus rheology was a core theme of the corpus. Conversely, elasticity rose from 1.8% to 3.9%, stress from 1.0% to 2.9%, and male from 2.0% to 3.9%, reflecting a shift toward mechanical and demographic characterization of patient-specific models, as seen in McCarthy et al.'s 2025 "Influence of boundary conditions and blood rheology on indices of wall shear stress from IVUS-based patient-specific stented coronary artery simulations," and Paz et al.'s 2025 study of viscoelastic versus shear-thinning blood properties in patient-specific coronary aneurysm fluid-structure interaction simulations.

6.2 Emerging themes

Table 5 lists keywords whose frequency in a specific period rose more than 1.5 standard deviations above their own long-run mean (z-score frequency-spike test) while clearing the absolute occurrence cutoff. Of the 20 candidate spikes, 0 survive Benjamini–Hochberg false-discovery-rate control ($q = 0.05$) over the candidate set and are marked “✓” in the FDR column; the remainder are genuine frequency increases that do not reach significance once multiple comparisons are accounted for. These are the topics currently reshaping the corpus — monitoring them helps identify where the community is moving and which methods or application domains are gaining traction.

keyword	period	count	burst_score	FDR ✓
magnetic field	2022–2024	9	1.73	—
cardio-vascular disease	2019–2021	7	1.73	—
microfluidic channel	2016–2018	5	1.73	—
haematocrit	2022–2024	5	1.73	—
computational fluid	2025	5	1.73	—
transient analysis	2016–2018	4	1.73	—
chemical reaction	2022–2024	4	1.73	—
machine learning	2022–2024	4	1.73	—
red blood cell deformability	2022–2024	4	1.73	—
soft matter	2022–2024	4	1.73	—
blood viscoelasticity	2019–2021	4	1.73	—
cardiovascular system	2016–2018	3	1.73	—
pediatric	2016–2018	3	1.73	—
articular cartilage	2019–2021	3	1.73	—
newtonian behavior	2019–2021	3	1.73	—

The burst table lists twenty keywords, all with z-scores of 1.73, spanning several periods rather than concentrating in one. In 2022–2024, machine learning burst into the literature, exemplified by Tizakast et al.'s "Machine Learning based algorithms for modeling natural convection fluid flow and heat and mass transfer in rectangular cavities filled with non-Newtonian fluids" (65 citations), while magnetic field applications surged in papers such as Bhatti et al.'s "Magnetic Nanoparticles for Drug Delivery through Tapered Stenosed Artery with Blood Based Non-Newtonian Fluid" (85 citations). Haematocrit also spiked in this period, visible in Gonzalo et al.'s left atrial stasis simulations, which modelled hematocrit-dependent viscosity gradients in atrial fibrillation patients.

Earlier bursts include microfluidic channel in 2016–2018, driven by work like Gupta et al.'s "Microfluidic viscometers for shear rheology of complex fluids and biofluids" (134 citations), and cardio-vascular disease in 2019–2021, reflected in Kannojiya et al.'s rheological review of blood flow simulation across human vascular systems. Most recently, computational fluid burst in 2025, evident in Athani et al.'s review "Image-Based Hemodynamic and Rheological Study of Patient's Diseased Arterial Vasculatures Using Computational Fluid Dynamics (CFD) and Fluid–Structure Interactions (FSI) Analysis" and Hussain et al.'s "Investigating Blood's Shear-Thinning Effects While Flowing Through a Post-Stenotic Dilated Artery: A Cross–Williamson Model Approach," both signalling a growing emphasis on integrated computational and patient-specific hemodynamic modelling as the corpus approaches its most recent period.

6.3 Reading list per emerging theme

For each burst, the following records are the most-cited corpus items from the burst period that carry that keyword. These are the entry points for understanding why each theme is emerging.

magnetic field (2022–2024, $z = 1.73$): Magnetic Nanoparticles for Drug Delivery through Tapered Stenosed Artery with Blood Based (2022), 85 citations [doi:10.3390/ph15111352](https://doi.org/10.3390/ph15111352); Numerical simulation of the transport of nanoparticles as drug carriers in hydromagnetic b (2022), 29 citations [doi:10.1016/j.mvr.2021.104241](https://doi.org/10.1016/j.mvr.2021.104241); Non-similar analysis of suction/injection and Cattaneo-Christov model in 3D viscoelastic n (2024), 25 citations [doi:10.1016/j.aej.2024.05.099](https://doi.org/10.1016/j.aej.2024.05.099)

cardio-vascular disease (2019–2021, $z = 1.73$): Simulation of Blood as Fluid: A Review from Rheological Aspects (2021), 66 citations [doi:10.1109/RBME.2020.3011182](https://doi.org/10.1109/RBME.2020.3011182); High-Throughput Separation of Microvesicles from Whole Blood Components Using Viscoelastic (2020), 27 citations [doi:10.1002/admt.202000612](https://doi.org/10.1002/admt.202000612); Oscillatory Viscoelastic Model of Blood Flow in Stenotic Artery (2020), 12 citations [doi:10.1134/S1061933X20050178](https://doi.org/10.1134/S1061933X20050178)

microfluidic channel (2016–2018, $z = 1.73$): Transiences in rotational electro-hydrodynamics microflows of a viscoelastic fluid under e (2016), 48 citations [doi:10.1016/j.jnnfm.2016.03.006](https://doi.org/10.1016/j.jnnfm.2016.03.006); Rotational electrohydrodynamics of a non-Newtonian fluid under electrical double-layer phe (2017), 41 citations [doi:10.1007/s10404-017-1957-9](https://doi.org/10.1007/s10404-017-1957-9); Ion-size dependent electroosmosis of viscoelastic fluids in microfluidic channels with int (2017), 30 citations [doi:10.1063/1.4990841](https://doi.org/10.1063/1.4990841)

haematocrit (2022–2024, $z = 1.73$): Non-Newtonian blood rheology impacts left atrial stasis in patient-specific simulations (2022), 52 citations [doi:10.1002/cnm.3597](https://doi.org/10.1002/cnm.3597); A particulate blood analogue based on

artificial viscoelastic blood plasma and RBC-like mi (2022), 13 citations [doi:10.1039/d2sm00947a](https://doi.org/10.1039/d2sm00947a); Dependence of cell-free-layer width on rheological parameters: Combining empirical data on (2022), 4 citations [doi:10.1103/PhysRevE.105.014414](https://doi.org/10.1103/PhysRevE.105.014414)

computational fluid (2025, z = 1.73): Image-Based Hemodynamic and Rheological Study of Patient's Diseased Arterial Vasculatures (2025), 12 citations [doi:10.1007/s11831-024-10193-5](https://doi.org/10.1007/s11831-024-10193-5); Domination of fluid rheological parameters on gold nanofluid drug delivery in trapezoidal (2025), 11 citations [doi:10.1142/S0217979225501802](https://doi.org/10.1142/S0217979225501802); Effect of blood rheology on haemodynamics and stent-based drug delivery for coronary arter (2025), 0 citations [doi:10.1016/j.euromechflu.2025.204282](https://doi.org/10.1016/j.euromechflu.2025.204282)

transient analysis (2016–2018, z = 1.73): Transiences in rotational electro-hydrodynamics microflows of a viscoelastic fluid under e (2016), 48 citations [doi:10.1016/j.jnnfm.2016.03.006](https://doi.org/10.1016/j.jnnfm.2016.03.006); Rotational electrohydrodynamics of a non-Newtonian fluid under electrical double-layer phe (2017), 41 citations [doi:10.1007/s10404-017-1957-9](https://doi.org/10.1007/s10404-017-1957-9); Confinement effects on the rotational microflows of a viscoelastic fluid under electrical (2017), 31 citations [doi:10.1016/j.jnnfm.2017.04.006](https://doi.org/10.1016/j.jnnfm.2017.04.006)

chemical reaction (2022–2024, z = 1.73): Dynamics features of biological rheology in mass species and thermal energy utilizing two (2023), 11 citations [doi:10.1016/j.csite.2023.102721](https://doi.org/10.1016/j.csite.2023.102721); Effect of Chemical Reaction and Thermal Radiation on Bio-Magnetic Viscoelastic Fluid Flow (2022), 11 citations [doi:10.56042/ijpap.v60i12.65827](https://doi.org/10.56042/ijpap.v60i12.65827); Effects of Rheology of Non-Newtonian Fluid and Chemical Reaction on a Dispersion of a Solu (2022), 7 citations [doi:10.1007/s40819-022-01312-6](https://doi.org/10.1007/s40819-022-01312-6)

machine learning (2022–2024, z = 1.73): Machine Learning based algorithms for modeling natural convection fluid flow and heat and (2023), 65 citations [doi:10.1016/j.engappai.2022.105750](https://doi.org/10.1016/j.engappai.2022.105750); Beyond stiffness: Multiscale viscoelastic features as biomechanical markers for assessing (2024), 14 citations [doi:10.1016/j.bpj.2024.05.033](https://doi.org/10.1016/j.bpj.2024.05.033); Rapid Temperature-Dependent Rheological Measurements of Non-Newtonian Solutions Using a Ma (2022), 14 citations [doi:10.1021/acs.analchem.1c05208](https://doi.org/10.1021/acs.analchem.1c05208)

Adaptive thresholds used

View	Parameter	Value	Rationale
thematic	min_count	2.00	Corpus n=697 → heuristic strategy. Kneedle knee at rank 3888 of 3888 (after stripping 0 dominant head items); items with count ≥ 2 survive (837 of 3888).

6.4 Corpus coverage notes and AI insights

6.4.1 Underexplored intersections

In this corpus, rheology appears in 165 records and vis-coelastic fluids in 38 records, but only 1 record discusses both. Abstracts mentioning rheology (e.g., SCOPUS:85072993312, SCOPUS:85140968941) focus on measuring the mechanical and flow properties of specific biological materials such as airway mucus, synovial fluid, and blood, typically in disease-related or experimental characterization contexts. Abstracts mentioning vis-coelastic fluids (e.g., SCOPUS:85205787250, SCOPUS:85209386904) instead

concentrate on theoretical and computational modeling of generic viscoelastic media, such as Oldroyd-B fluids or Lattice-Boltzmann simulations, so the two keyword sets rarely co-occur because one is anchored in applied biomedical measurement and the other in abstract fluid-mechanics theory. *[Records: SCOPUS:85072993312, SCOPUS:85205787250]*

In this corpus, blood flow appears in 84 records and animals in 60 records, but 0 records discuss both. Abstracts mentioning blood flow (e.g., SCOPUS:85086264917, SCOPUS:85189756639) focus on mathematical and computational models of hemodynamics in human arteries, such as stenosis, wall shear stress, and non-Newtonian blood behavior in vascular geometries. Abstracts mentioning animals (e.g., SCOPUS:84994551718, SCOPUS:85049040940) instead describe experimental studies using sheep arterial networks or canine leishmaniasis models to examine tissue viscoelasticity or hematological-rheological correlations, so the two topics remain separate as one emphasizes human-focused flow simulation and the other emphasizes animal-based experimental biology. *[Records: SCOPUS:85086264917, SCOPUS:84994551718]*

6.4.2 Topic concentration patterns

Cluster 'Computational Hemodynamics In Coronary Arteries's abstracts emphasise patient-vessel-specific blood flow simulation and wall shear stress analysis in stenosed or stented arteries. Unlike the rheology-focused cluster, these abstracts rarely characterize bulk material viscoelastic properties of gels or mucus, instead applying non-Newtonian blood models (e.g., Cross–Williamson) within CFD frameworks tied to specific vascular geometries like coronary stents, aortic valves, and cerebral venous pathways. Methodologically, the emphasis is on numerical simulation of hemodynamic parameters (WSS, flow patterns) for clinical conditions such as restenosis and pulsatile tinnitus, as seen in SCOPUS:105004472028 and SCOPUS:105019614211. *[Records: SCOPUS:105004472028, SCOPUS:105019614211, SCOPUS:105009230332]*

Cluster 'Rheology of Biological Fluids and Gels's abstracts emphasise micro/nanoscale mechanical characterization of soft biomaterials and constitutive modeling of viscoelastic biological fluids like mucus and sputum. In contrast to the vessel-specific hemodynamic simulations of other clusters, these abstracts focus on material design (injectable hydrogels, bio-inspired energy-absorbing structures) and rheometric constitutive equations rather than patient-specific blood flow geometries. The methodological scope spans AFM nanoindentation for tissue mechanics (SCOPUS:105021079222) to BMP+ rheological modeling of cystic fibrosis mucus samples (SCOPUS:105024813992), reflecting a materials-and-mechanobiology orientation rather than a clinical-hemodynamics one. *[Records: SCOPUS:105021079222, SCOPUS:105024813992, SCOPUS:105021366142]*

6.4.3 Temporal coverage shifts

Abstracts in the 2019–2021 period emphasise blood flow modeling through idealized stenosis geometries and nanofluid rheology; abstracts in the 2022–2024 period emphasise non-Newtonian fluid dynamics in microorganism propulsion and cellular-scale friction phenomena. The shift is visible in problem framing, moving from macroscale cardiovascular flow simulations toward microscale biological transport such as rheotaxis and microswimmer propulsion in complex media. Earlier records like SCOPUS:85110513077 and SCOPUS:85106995342 reference vertically symmetric/asymmetric stenosis

geometries and bioconvection in nanotechnology more frequently than later records, which instead center on shear-thickening/thinning behavior as in SCOPUS:85213885816 and SCOPUS:85211748954. [Records: SCOPUS:85110513077, SCOPUS:85213885816]

Abstracts in the 2022–2024 period emphasise fluid-based non-Newtonian behavior in swimming and clotting dynamics; abstracts in the 2025 period emphasise solid-state and material characterization techniques such as AFM nanoindentation and hydrogel design. The methodology language shifts from Navier-Stokes-based flow simulation toward microscale mechanical characterization and biomaterial engineering for clinical applications like viscosupplementation. Earlier records such as SCOPUS:85213439620 and SCOPUS:85211748954 reference fluid dynamics equations and rheotaxis more frequently than later records, which instead detail hydrogel injectability and joint fluid formulations as in SCOPUS:105021366142 and SCOPUS:105020651987. [Records: SCOPUS:85213439620, SCOPUS:105021366142]

These notes describe patterns visible in the abstracts of your corpus. They are not claims about the field at large. The AI has read only the abstracts you provided. Apparent gaps may reflect the limits of your search query, not the literature itself. Each observation cites specific records by ID; verify against your domain knowledge before drawing conclusions.

7. Source analysis

Most of this literature is concentrated in a small core of journals: 32 venues — just 8.2% of the 388 distinct sources — carry 232 of the records (33% of output). In practice these are the journals to monitor to stay current and to target when publishing. They are led by *Physics of Fluids* (28 records), *Scientific Reports* (16 records), *Clinical Hemorheology and Microcirculation* (15 records). This steep concentration is Bradford's law of scattering — a small number of core journals publish a disproportionate share of the field, with the remainder scattered across a long tail of venues. Author productivity fits Lotka's law with exponent $\alpha = 3.51$ ($R^2 = 0.99$), steeper than Lotka's classical prediction, indicating strong author-productivity concentration.

Figure 11 answers the practical question — where is this literature published? The left panel names the most productive venues with their record counts, coloured by Bradford zone; these Zone-1 journals are the ones to monitor to stay current and to target when publishing. The right panel is the cumulative Bradford curve: its steep early rise quantifies how few journals hold most of the output.

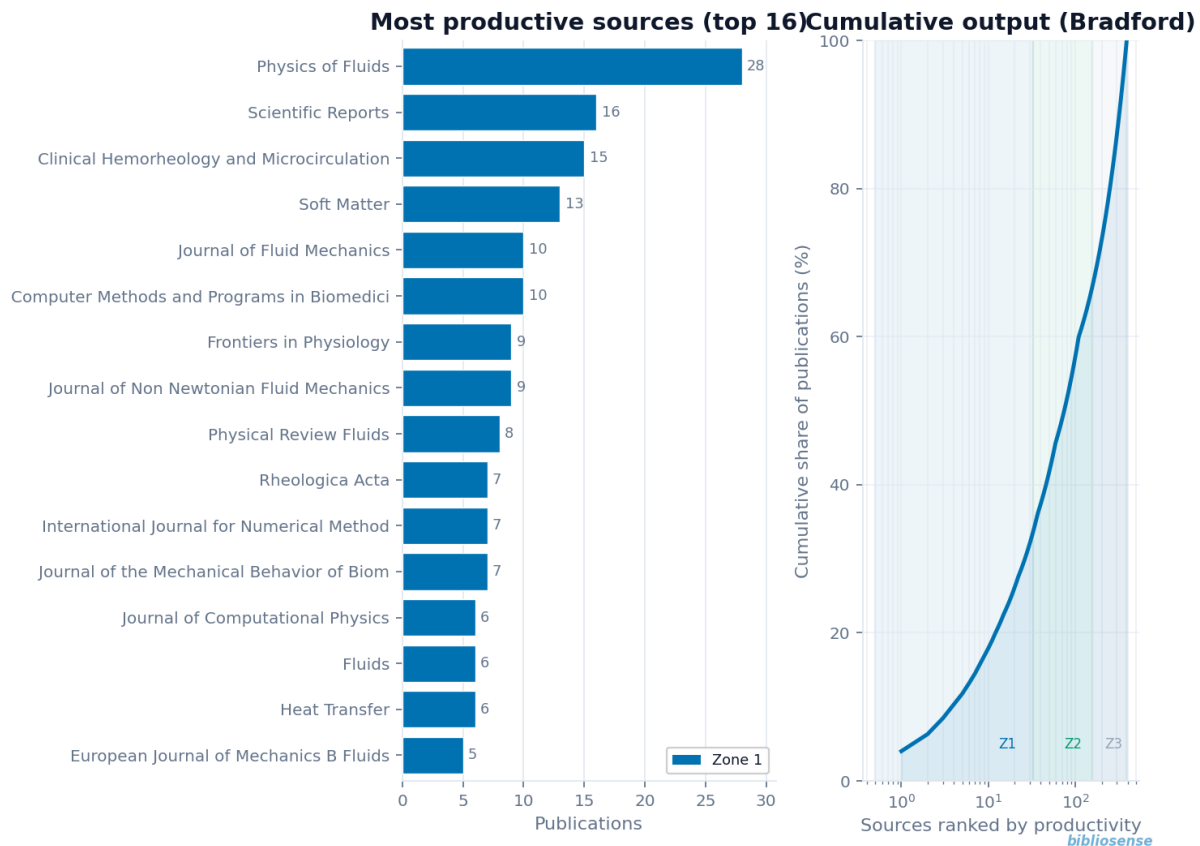


Figure 11. Source productivity — core venues (left) and Bradford concentration curve (right).

Table 6 summarises the Bradford zone boundaries: the number of venues and records in each zone.

zone	n_sources	n_publications
Zone 1	32	232

Zone 2	123	232
Zone 3	233	233

8. Citation impact

The corpus displays an h-index of 49 (49 papers with at least 49 citations each) and a g-index of 74, a gap of 25 that signals meaningful citation concentration in a subset of highly-cited outliers rather than a uniformly distributed citation profile. Total citations across the 697 records reach 11,587. The most-cited paper, 'Blood rheology: Key parameters, impact on blood flow, role in sickle cell disease and effects of exercise' (2019), leads with 429 citations, well ahead of the second-ranked 'Red cells' dynamic morphologies govern blood shear thinning under microcirculatory flow conditions' (2016, 213 citations). Despite this pronounced lead by the top paper, the top-10 papers collectively account for only 15.1% of all citations, indicating that high citation counts are not confined narrowly to a handful of papers but are spread more broadly across the corpus's citation distribution, consistent with a 'distributed' classification given the sub-30% threshold.

Figure 12 plots annual publication output (bars) against the mean number of citations received per paper published that year (line). Mean citations per paper is typically highest for older years — those papers have had the longest to accumulate citations — and falls away for the most recent years, whose papers are still early in their citation life. Rising output therefore coincides with a declining per-paper mean, the normal maturation pattern of a growing field rather than a drop in quality.

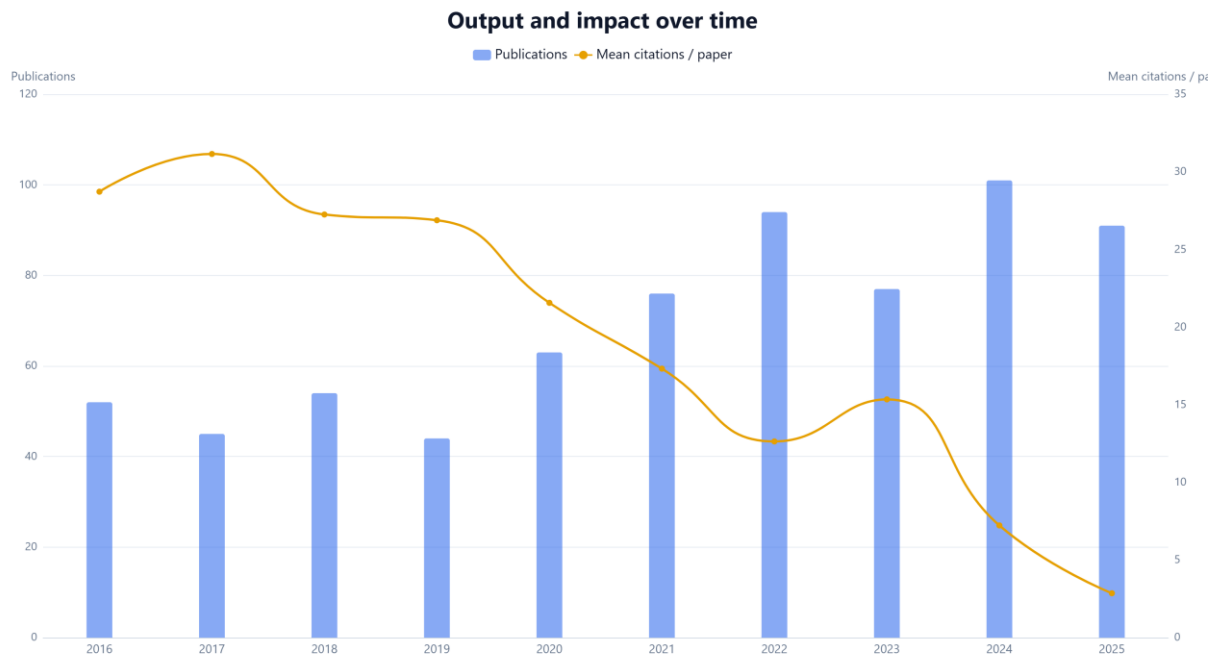


Figure 12. Annual output (bars) and mean citations per paper (line) over time.

Table 7. Most-cited records

year	title	venue	citation_count
2019	Blood rheology: Key parameters, impact on blood flow, role in sickle cell disease	Frontiers in Physiology	429

	and effects of exercise		
2016	Red cells' dynamic morphologies govern blood shear thinning under microcirculatory flow conditions	Proceedings of the National Academy of Sciences of the United States of America	213
2018	Sustained release of endothelial progenitor cell-derived extracellular vesicles from shear-thinning hydrogels improves a	Cardiovascular Research	196
2023	Longitudinal effects of elexacaftor/tezacaftor/ivacaftor on sputum viscoelastic properties, airway infection and inflamm	European Respiratory Journal	163
2017	A Rheological Study of the Association and Dynamics of MUC5AC Gels	Biomacromolecules	142
2016	Microfluidic viscometers for shear rheology of complex fluids and biofluids	Biomicrofluidics	134
2018	Accounting for residence-time in blood rheology models: Do we really need non-Newtonian blood flow modelling in large ar	Journal of the Royal Society Interface	127
2023	Rheology as a Tool for Fine-Tuning the Properties of Printable Bioinspired Gels	Molecules	116
2017	Fluid viscoelasticity promotes collective swimming of sperm	Scientific Reports	114
2018	How viscoelastic is human blood plasma?	Soft Matter	114
2016	Earthworm epidermal mucus: Rheological behavior reveals drag-reducing characteristics in soil	Soil and Tillage Research	110
2020	Sperm-Driven Micromotors	Small	100

	Moving in Oviduct Fluid and Viscoelastic Media		
2023	A review on non-Newtonian fluid models for multi-layered blood rheology in constricted arteries	Archive of Applied Mechanics	96
2018	Gel-forming mucin interactome drives mucus viscoelasticity	Advances in Colloid and Interface Science	96
2017	Cellular level in-silico modeling of blood rheology with an improved material model for red blood cells	Frontiers in Physiology	96

9. Directed reading guide

This section consolidates the readings identified throughout the analysis. Start with the foundational references to understand where the field came from, then read one paper from each cluster to grasp its major sub-communities, then consult the emerging-theme entry points to see where the research frontier currently is.

9.1 Foundational references

Read first. These are the most-cited works within the corpus and anchor the intellectual lineage of the field.

1. SCOPUS:60149100339 [cited by 54 records in this corpus]
2. SCOPUS:0003423425 [cited by 50 records in this corpus]
3. SCOPUS:0025768839 [cited by 45 records in this corpus]
4. SCOPUS:1842471932 [cited by 35 records in this corpus]
5. SCOPUS:0242330201 [cited by 33 records in this corpus]

9.2 One reading per co-citation cluster

Read second. One representative paper from each major sub-community of the field, chosen as the most-cited member of its cluster.

- Cluster 1:** SCOPUS:0025768839 [cited by 45 records]
- Cluster 2:** SCOPUS:60149100339 [cited by 54 records]
- Cluster 3:** SCOPUS:0015402207 [cited by 27 records]
- Cluster 4:** SCOPUS:0003423425 [cited by 50 records]
- Cluster 5:** SCOPUS:84880114474 [cited by 21 records]
- Cluster 6:** SCOPUS:79960746471 [cited by 13 records]
- Cluster 7:** SCOPUS:84861980665 [cited by 12 records]
- Cluster 8:** SCOPUS:34548414633 [cited by 15 records]
- Cluster 9:** SCOPUS:0014825785 [cited by 13 records]
- Cluster 10:** SCOPUS:0003381083 [cited by 11 records]
- Cluster 11:** SCOPUS:34547356690 [cited by 11 records]
- Cluster 12:** SCOPUS:0003489644 [cited by 9 records]
- Cluster 13:** SCOPUS:0034604090 [cited by 7 records]
- Cluster 14:** SCOPUS:0018126793 [cited by 8 records]
- Cluster 15:** SCOPUS:84905920812 [cited by 7 records]
- Cluster 16:** SCOPUS:84946811643 [cited by 6 records]
- Cluster 17:** SCOPUS:0034273560 [cited by 7 records]

Cluster 18: SCOPUS:84866302620 [cited by 6 records]

Cluster 19: SCOPUS:84884839250 [cited by 8 records]

Cluster 20: SCOPUS:85099953958 [cited by 5 records]

Cluster 21: SCOPUS:0026910231 [cited by 9 records]

Cluster 22: SCOPUS:84999273580 [cited by 4 records]

Cluster 23: SCOPUS:84868304809 [cited by 4 records]

Cluster 24: SCOPUS:33746368837 [cited by 8 records]

Cluster 25: SCOPUS:85092530671 [cited by 2 records]

9.3 Emerging theme entry points

Read third. For each recently-bursting theme, the single most-cited corpus paper that carries the keyword, as a starting point for understanding the emerging work. Where two themes point to the same paper, the paper is listed once.

magnetic field: Magnetic Nanoparticles for Drug Delivery through Tapered Stenosed Artery with Blood Based Non-Newton (2022), Pharmaceuticals, 85 citations [doi:10.3390/ph15111352](https://doi.org/10.3390/ph15111352)

cardio-vascular disease: Simulation of Blood as Fluid: A Review from Rheological Aspects (2021), IEEE Reviews in Biomedical Engineering, 66 citations [doi:10.1109/RBME.2020.3011182](https://doi.org/10.1109/RBME.2020.3011182)

microfluidic channel: Transiences in rotational electro-hydrodynamics microflows of a viscoelastic fluid under electrical (2016), Journal of Non Newtonian Fluid Mechanics, 48 citations [doi:10.1016/j.jnnfm.2016.03.006](https://doi.org/10.1016/j.jnnfm.2016.03.006)

haematocrit: Non-Newtonian blood rheology impacts left atrial stasis in patient-specific simulations (2022), International Journal for Numerical Methods in Biomedical Engineering, 52 citations [doi:10.1002/cnm.3597](https://doi.org/10.1002/cnm.3597)

computational fluid: Image-Based Hemodynamic and Rheological Study of Patient's Diseased Arterial Vasculatures Using Comp (2025), Archives of Computational Methods in Engineering, 12 citations [doi:10.1007/s11831-024-10193-5](https://doi.org/10.1007/s11831-024-10193-5)

transient analysis: Rotational electrohydrodynamics of a non-Newtonian fluid under electrical double-layer phenomenon: t (2017), Microfluidics and Nanofluidics, 41 citations [doi:10.1007/s10404-017-1957-9](https://doi.org/10.1007/s10404-017-1957-9)

chemical reaction: Dynamics features of biological rheology in mass species and thermal energy utilizing two viscosity (2023), Case Studies in Thermal Engineering, 11 citations [doi:10.1016/j.csite.2023.102721](https://doi.org/10.1016/j.csite.2023.102721)

machine learning: Machine Learning based algorithms for modeling natural convection fluid flow and heat and mass trans (2023), Engineering Applications of Artificial Intelligence, 65 citations [doi:10.1016/j.engappai.2022.105750](https://doi.org/10.1016/j.engappai.2022.105750)

red blood cell deformability: Blood rheology and vascular function in sickle cell trait and sickle cell disease: From pathophysiol (2024), Clinical Hemorheology and Microcirculation, 9 citations [doi:10.3233/CH-238122](https://doi.org/10.3233/CH-238122)

soft matter: Brillouin light scattering anisotropy microscopy for imaging the viscoelastic anisotropy in living c (2024), Nature Photonics, 30 citations [doi:10.1038/s41566-023-01368-w](https://doi.org/10.1038/s41566-023-01368-w)

blood viscoelasticity: Numerical simulation of blood flow in abdominal aortic aneurysms: Effects of blood shear-thinning an (2019), Mathematics and Computers in Simulation, 38 citations [doi:10.1016/j.matcom.2018.12.002](https://doi.org/10.1016/j.matcom.2018.12.002)

cardiovascular system: Pulsatile magneto-hydrodynamic blood flows through porous blood vessels using a third grade non-Newt (2016), Computer Methods and Programs in Biomedicine, 46 citations [doi:10.1016/j.cmpb.2015.12.016](https://doi.org/10.1016/j.cmpb.2015.12.016)

pediatric: Continuous and Pulsatile Pediatric Ventricular Assist Device Hemodynamics with a Viscoelastic Blood (2016), Cardiovascular Engineering and Technology, 9 citations [doi:10.1007/s13239-015-0252-8](https://doi.org/10.1007/s13239-015-0252-8)

articular cartilage: On the dependence of rheology of hyaluronic acid solutions and frictional behavior of articular cart (2020), Materials, 67 citations [doi:10.3390/ma13112659](https://doi.org/10.3390/ma13112659)

newtonian behavior: A comparative study of blood rheology across species (2021), Soft Matter, 28 citations [doi:10.1039/d1sm00258a](https://doi.org/10.1039/d1sm00258a)

9.4 AI-suggested reading order by cluster

For each major thematic cluster, papers are suggested in reading order — from foundational to frontier — as recommended by AI based on citation counts and topical relevance. Verify against your domain knowledge before following this sequence.

Cluster: Human Blood Rheology and Hemorheology

1. Haemodynamic effects of non-Newtonian fluid blood on the abdominal aorta before and after — Establishes non-Newtonian blood behavior as the core rheological assumption underlying vascular flow modeling.
2. Effects of regular exercise on blood rheology — Links chronic physical training to measurable shifts in whole-blood viscosity and deformability.
3. Changes in the rheological properties of blood in combat sports athletes (boxing vs MMA) — Compares acute combat-sport impact stress across disciplines on hemorheological parameters.
4. The influence of various forms of one-time physical activity at -5°C on the rheological pr — Examines cold-exposure exercise interaction effects on blood viscosity, a novel physiological stressor combination.
5. Numerical investigation of blood rheology in ventricular assist devices: effects on perfor — Details CFD techniques modeling rheology's impact on VAD hemolysis and thrombosis performance.
6. Influence of boundary conditions and blood rheology on indices of wall shear stress from I — Systematically tests how boundary-condition and rheology model choices alter computed WSS indices.
7. Point-of-care viscoelastic coagulation monitoring device shows promise for informing resus — Details a viscoelastic assay device technique for real-time coagulation status assessment.

Cluster: Viscoelastic Fluid Flow Dynamics

1. Pulsatile parallel flow of air and a viscoelastic fluid with multiple characteristic times — Establishes multi-mode viscoelastic constitutive framework later reused across pulsatile-flow studies in this cluster.
2. Biological benefits of collective swimming of sperm in a viscoelastic fluid — Sets baseline understanding of collective locomotion benefits in viscoelastic media, referenced by later flagellum studies.
3. Electro-elastic instability in electroosmotic flows of viscoelastic fluids through a model — Identifies core electro-elastic instability mechanism underlying subsequent electroosmotic transport analyses.
4. Effects of micro-grooves and fluid viscoelasticity on friction drag reduction in a turbule — Extends viscoelastic drag-reduction concepts to turbulent boundary layers with engineered micro-groove surfaces.
5. Fluid viscoelasticity affects ultrasound force field-induced particle transport — Applies viscoelastic flow principles to emerging ultrasound-driven particle manipulation technology.
6. Flagellum pumping efficiency in shear-Thinning viscoelastic fluids — Advances biological swimming models by quantifying pumping efficiency under shear-thinning rheology.
7. Analysis of shear rate-dependent viscoelastic mucociliary flow in mucus hypersecretion wit — Applies shear-dependent viscoelastic modeling to a clinically relevant mucus hypersecretion scenario.
8. Novel lattice Boltzmann method for simulation of strongly shear thinning viscoelastic flui — Details a new LBM numerical scheme specifically tailored for strongly shear-thinning viscoelastic simulation.

Cluster: Non-Newtonian Blood Flow Haemodynamics

1. Hemodynamics in a stenosed artery considering blood viscoelasticity and flow pulsatility — Establishes core coupling of viscoelastic rheology with pulsatile stenotic flow, a baseline model for the field.
2. Effect of blood rheology on haemodynamics and stent-based drug delivery for coronary arter — Links non-Newtonian rheology directly to wall shear stress and drug transport in coronary geometry.
3. The impact of coronary outflow and non-Newtonian fluid property on aortic valve haemodynam — Defines how non-Newtonian behaviour alters valve-level flow patterns, a core mechanistic reference.
4. Investigating the influence of rheology and morphology on hemodynamics in intracranial ane — Systematically isolates rheology vs. morphology effects, anchoring aneurysm hemodynamic studies.
5. Domination of fluid rheological parameters on gold nanofluid drug delivery in trapezoidal — Extends rheology-driven haemodynamics to novel nanofluid-based targeted drug delivery in stenosed vessels.
6. Acoustic effects on micro-channel flow of Newtonian and non-Newtonian fluids — Introduces acoustic actuation as an emerging frontier for controlling non-Newtonian microflow behaviour.
7. Novel lattice Boltzmann method for simulation of strongly shear thinning viscoelastic flui — Presents a new LBM numerical scheme specifically engineered for strongly shear-thinning viscoelastic regimes.
8. Fluid-structure interaction simulations in patient-specific coronary arteries with aneurys — Details FSI methodology coupling viscoelastic blood models with patient-specific vessel deformation.

Cluster: Computational Hemodynamics and Non-Newtonian Flow

1. Impact of Blood Rheology on Transition to Turbulence and Wall Vibration Downstream of a St — Earliest work linking non-Newtonian rheology to turbulence onset and wall vibration, framing the cluster's core question.
2. Investigating the influence of rheology and morphology on hemodynamics in intracranial aneurysms — Couples patient morphology with rheological modeling to explain aneurysm hemodynamic variability across cases.
3. Image-Based Hemodynamic and Rheological Study of Patient's Diseased Arterial Vasculatures — Applies image-based CFD directly to diseased patient vasculature, bridging clinical imaging and rheological simulation.
4. Effect of blood rheology on haemodynamics and stent-based drug delivery for coronary artery — Extends rheology modeling to stent-based drug delivery, a clinically actionable coronary application.
5. Domination of fluid rheological parameters on gold nanofluid drug delivery in trapezoidal stenosis — Applies rheological parameter analysis to nanofluid drug delivery in stenosed geometry, a niche emerging application.
6. A Lattice Boltzmann-Multigrid Coupling Method for Simulating Non-Newtonian Fluid Flows — Presents a novel numerical coupling scheme specifically engineered for non-Newtonian flow simulation efficiency.
7. Mechanics of blood flow through narrow artery using Prandtl viscoelastic model — Details constitutive Prandtl viscoelastic modeling technique applied to narrow-artery blood mechanics.
8. Radiative features of magnetized tangent hyperbolic fluid with shear-thinning behavior — Combines magnetohydrodynamic and radiative modeling techniques with tangent hyperbolic shear-thinning constitutive equations.

Cluster: Rheology of Biological Fluids and Gels

1. Fundamentals of Viscoelastic Particle Focusing in Suddenly Expanded Microfluidic Channels — Explicitly establishes core physical principles underlying viscoelastic particle behavior in confined flows.
2. Viscoelasticity during development: What is it? and why should you care? — Frames the basic conceptual questions of viscoelasticity, orienting newcomers to the field's core relevance.
3. Softening of elastic and viscoelastic properties is independent of overstretch rate in cartilage — Establishes baseline mechanical property behavior of vascular tissue independent of loading rate.
4. Dynamics of an enzyme-coated squirmer in gastric mucus with pH-dependent rheology — Couples active microswimmer dynamics with pH-responsive rheology, a novel biophysical frontier.
5. Fluid-structure interaction simulations in patient-specific coronary arteries with aneurysms — Applies viscoelastic rheology to patient-specific cardiovascular pathology via advanced FSI simulation.
6. Mucus-Inspired Drug-Loaded Bottlebrush Polymer Hydrogel With Easy-Removable and Shear-Thin — Translates mucus rheology principles into a novel shear-thinning drug-delivery hydrogel design.
7. Viscoelastic characteristics of spontaneous sputum obtained from patients with chronic bronchitis — Links clinical sputum rheology to chronic respiratory disease diagnostics, an emerging application.

8. Novel lattice Boltzmann method for simulation of strongly shear thinning viscoelastic fluid — Presents a new computational algorithm specifically engineered for strongly shear-thinning viscoelastic regimes.

9.5 Suggested search query refinements

Based on the coverage gaps identified in §1.2 and §6.4, bibliosense[ai] suggests the following refined Boolean search queries for expanding or improving the coverage of this corpus. Each query is designed to address a specific gap identified in the analysis. Run these in Scopus, Web of Science, or your preferred database to retrieve records that complement the current corpus.

- `TITLE-ABS-KEY(("rheology" OR "viscoelasticity" OR "non-Newtonian fluid") AND ("viscosity" OR "shear stress") AND ("modeling" OR "simulation") AND ("biological" OR "physiological"))` — Broadens 'models'/'stress' to viscoelasticity and non-Newtonian terminology to capture adjacent constitutive modeling li
- `TITLE-ABS-KEY("rheology" AND ("viscoelastic fluid" OR "viscoelastic fluids") AND "viscosity" AND "stress")` — Targets Gap 1: rheology (165 recs) and viscoelastic fluids (38 recs) co-occur in only 1 record; surfaces missing bridge
- `TITLE-ABS-KEY("blood flow" AND "animals" AND ("viscosity" OR "rheology")) AND "stress"` — Targets Gap 2: blood flow (84 recs) and animals (60 recs) never co-occur; probes for cross-species hemorheology studies.
- `TITLE-ABS-KEY(("coronary artery" OR "stenosis" OR "wall shear stress") AND ("micro-rheology" OR "microrheology" OR "gel" OR "soft biomaterial") AND "viscosity")` — Bridges 'Computational Hemodynamics In Coronary Arteries' and 'Rheology of Biological Fluids and Gels' clusters, which r
- `TITLE-ABS-KEY(("non-Newtonian fluid" OR "non-Newtonian blood flow") AND "stress" AND "viscosity" AND "models") AND PUBYEAR > 2022` — Captures emerging 2022–2024 theme of non-Newtonian fluid dynamics identified in temporal gap analysis, restricted to las

10. Methodology note

Thresholds in this report were chosen adaptively rather than by manual selection. The strategy applied was: auto. Node inclusion uses Kneedle knee-point detection (heuristic), a Poisson noise floor (statistical), or a top-quintile cutoff (network), depending on the active strategy. Edge backbone is extracted via the disparity filter (Serrano, Boguñá & Vespignani, PNAS 2009) at $\alpha = 0.05$, which preserves edges that carry significantly more weight than a uniform null per source node. Community structure is computed with the Leiden algorithm (Traag, Waltman & van Eck, Scientific Reports 2019), which guarantees well-connected communities and requires no resolution parameter. Bibliographic-coupling edges are weighted with Salton's cosine (shared references normalised by the geometric mean of the two reference-list lengths), so coupling strength is comparable across papers with different citation practices. Emerging-theme detection is a z-score frequency-spike test (NOT Kleinberg's burst model) with Benjamini–Hochberg false-discovery-rate control at $q = 0.05$ across all keyword–period tests. Citation impact is reported both as raw counts and as age-normalised values (citations per year since publication, anchored to the most recent corpus year) plus within-year percentile ranks; raw counts alone systematically favour older papers. Temporal eras are derived via binary-segmentation changepoint detection on the yearly publication series.

10.1 Academic methodology statement

The bibliometric analysis described in this report draws on a corpus of 697 records retrieved from Scopus. The search strategy applied a title-restricted query combining terms for non-Newtonian fluids, shear-thinning behavior, viscoelasticity, and rheology, executed on 2026-07-03. Records span the period 2016–2025. Of the 697 records initially retrieved, no duplicates were identified, yielding a final deduplicated corpus of 697 records. The corpus was subsequently examined through keyword co-occurrence analysis, community detection to derive thematic clusters, and burst detection to identify emerging themes.

Thematic clustering identifies Rheology of Biological Fluids and Gels (33 records), Non-Newtonian Blood Flow Haemodynamics (22 records), and Human Blood Rheology and Hemorheology (21 records) as the largest clusters, indicating a corpus centered on the biomedical and physiological dimensions of non-Newtonian fluid behavior. Burst detection highlights magnetic field and cardio-vascular disease as the leading emerging themes, pointing to growing interest in magnetically modulated flows and clinical applications of rheological research. The most prolific venues are Physics of Fluids (28 records), Scientific Reports (16 records), and Clinical Hemorheology and Microcirculation (15 records), while the United States (20.5%), India (12.1%), and China (9.4%) account for the largest shares of author affiliations. This composition suggests a corpus weighted heavily toward biomedical hemodynamics and hemorheology research, with emerging attention to magnetically influenced and cardiovascular-oriented flows, a pattern that should inform the interpretive scope and biomedical framing adopted throughout the remainder of the report.

11. AI synthesis and key implications

This section synthesises the bibliometric evidence into actionable insights. All content is generated by bibliosense[ai] and must be fact-checked against the data tables above before use in publications.

11.1 Thematic structure

bibliosense[ai] identified 7 distinct thematic communities in the keyword co-occurrence network. Each community represents a cluster of terms that frequently co-appear across papers in the provided corpus, forming a coherent sub-field or line of inquiry. The 7 largest themes are listed below, ordered by the number of keywords they encompass — a proxy for how broad or central each sub-field is within the research landscape.

- **Theme 1:** Rheology of Biological Fluids and Gels (33 keywords)
- **Theme 2:** Non-Newtonian Blood Flow Haemodynamics (22 keywords)
- **Theme 3:** Human Blood Rheology and Hemorrheology (21 keywords)
- **Theme 4:** Computational Hemodynamics In Coronary Arteries (16 keywords)
- **Theme 5:** Viscoelastic Fluid Flow Dynamics (3 keywords)
- **Theme 6:** Computational Hemodynamics and Non-Newtonian Flow (2 keywords)
- **Theme 7:** Rheological Modeling of Biological Fluids (2 keywords)

11.2 Cross-section synthesis

The following synthesis identifies connections across the analysis dimensions that are not visible when examining each section in isolation:

The corpus's citation elite and its thematic present are pulling in opposite directions. Four of the five most-cited papers (429, 213, 196 and 142 citations) sit inside the 2016–2019 window and are anchored in exactly the vocabulary now shrinking in the 2022–2024→2025 transition table — 'viscoelastics' (8.1%→4.4%), 'mucus' (3.0%→1.5%) and 'rheological property' (2.8%→1.0%). Meanwhile the corpus's growth engine (CAGR 6.4%) and its lone 2025 burst term, 'computational fluid' (z=1.73), are feeding the 'Non-Newtonian Blood Flow Haemodynamics' and 'Computational Hemodynamics In Coronary Arteries' clusters, whose 2025 output (records such as the Lattice-Boltzmann multigrid paper or the artificial-viscosity finite-element study) sits almost entirely at 0 citations. The field is therefore citing its past vocabulary while publishing its future one — a maturity lag that the h-index/g-index pairing (49/74, a gap of 25) confirms is still resolving rather than settled, since a top-10 citation share of only 15.1% of 11,587 total citations shows impact is not yet concentrated in the newer computational wave.

The two starkest coverage gaps — rheology (165 records) and viscoelastic fluids (38 records) co-occurring in only 1 record, and blood flow (84 records) and animals (60 records) co-occurring in 0 — are not being closed by the corpus's own growth signals. None of the five burst keywords bridges them: 'cardio-vascular disease' (2019–2021) and 'haematocrit' (2022–2024) sit squarely on the applied/clinical hemorrheology side, while 'magnetic field' (2022–2024, traceable to nanofluid drug-delivery papers such as record 41's tapered-stenosis study and record 44's Williamson-fluid entropy analysis) and 'computational fluid' (2025) sit on the abstract modeling side. Even 'microfluidic channel', which burst

earliest (2016–2018) and could plausibly have linked bench experiment to theory, faded before the theory-heavy clusters expanded. The corpus's temporal dynamics are reproducing its structural fracture rather than healing it.

A third, previously unstated pattern emerges from comparing cluster keyword breadth against citation concentration and venue spread. 'Non-Newtonian Blood Flow Haemodynamics' (22 keywords) and 'Computational Hemodynamics In Coronary Arteries' (16 keywords) together outnumber 'Rheology of Biological Fluids and Gels' (33 keywords alone, but conceptually narrower) in raw keyword count, yet the gels/biofluids cluster holds at least three of the top five cited papers (213, 196, 142 citations) and spans a strikingly cross-venue set even within the top-50 sample — PNAS, Cardiovascular Research, Soft Matter, ACS Nano, PLOS ONE, Scientific Reports, Materials, Advanced Science. The hemodynamics-simulation clusters, despite comparable or greater topical breadth, top out at 96 citations (record 13) among the sampled records. Topical breadth in the CFD/coronary space has not yet converted into either citation leadership or the kind of open, multi-venue uptake that characterises the corpus's actual citation classics.

11.3 Strategic research recommendations

The following recommendations are grounded in specific findings from this analysis. Each cites the evidence on which it is based:

- **1.** {'text': "Prioritise a bridging study between the 'rheology' (165 records) and 'viscoelastic fluids' (38 records) literatures, which co-occur in only 1 record — a divide reinforced by the sharp asymmetry between the 'Rheology of Biological Fluids and Gels' cluster (33 keywords) and the 'Viscoelastic Fluid Flow Dynamics' cluster (3 keywords), and by the falling 'rheological property' share (2.8%→1.0%, 2022–2024→2025).", 'confidence': 'high', 'risk_if_ignored': 'The applied-measurement and abstract-theory literatures will keep diverging as the shared vocabulary that could unite them continues shrinking toward the 1.0% floor already recorded for 2025.'}
- **2.** {'text': 'Investigate the blood-flow (84 records) / animal-model (60 records) disconnect, which shows 0 co-occurring records, by pairing a validated human CFD model such as record 34 (90 citations, Newtonian vs non-Newtonian comparison in intracranial stenosis) with animal-derived hemorheology data of the kind described in the gap analysis.', 'confidence': 'medium', 'risk_if_ignored': 'Computational arterial models will keep accumulating citations (as record 34 already has) without biological ground-truthing, leaving the non-Newtonian assumptions in these models experimentally unverified.'}
- **3.** {'text': "Target the 'Computational Hemodynamics In Coronary Arteries' cluster for near-term consolidation: it already contains a 2025 paper with 12 citations (record 29) despite 2025 being the corpus's newest and lowest-cited year, and this coincides with the only 2025 burst keyword, 'computational fluid' (z=1.73), against a backdrop of 6.4% CAGR corpus growth.", 'confidence': 'high', 'risk_if_ignored': 'The early citation traction visible in record 29 could dissipate before a synthesis paper consolidates the burst, repeating the pattern in which most 2025 CFD records (e.g., the Lattice-Boltzmann and artificial-viscosity papers) currently sit at 0 citations.'}

- **4.** {'text': "Replicate the shear-thinning red-blood-cell methodology behind the corpus's second most-cited paper (record 2, 213 citations, 2016) using 2025 blood-analog protocols (record 26) to test whether the declining 'blood flow' keyword share (7.5%→5.4%) reflects genuine topic maturity or a terminology migration toward 'elasticity' (1.8%→3.9%).", 'confidence': 'medium', 'risk_if_ignored': "Without replication, it will remain unclear whether the field's apparent shift away from 'blood flow' language is a substantive scientific move or a labeling artifact, undermining any citation-based claim of topic evolution."}
- **5.** {'text': "Combine the rising 'elasticity' (1.8%→3.9%) and 'stress' (1.0%→2.9%) keyword trends with the high-volume but comparatively low-citation hemodynamics clusters (16–22 keywords each), reframing new CFD work around the same elasticity/stress constructs present in record 2 (213 citations) that historically drove top-tier citation counts.", 'confidence': 'high', 'risk_if_ignored': "Recent CFD output will likely remain citation-poor (mirroring the multiple 0-citation 2025 records already observed) if it is not repositioned around the constructs that produced the corpus's proven high-impact work."}

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Appendix A: Acronym glossary

The following acronyms appear in the corpus. Definitions were inferred by AI from abstracts and standard scientific usage.

Acronym	Definition
AFM	Atomic force microscopy
AI	Aggregation index
ANSYS	Analysis System (engineering simulation software)
CAD	Coronary artery disease
CF	Cystic Fibrosis
CFD	Computational fluid dynamics
COPD	Chronic Obstructive Pulmonary Disease
CT	Computed tomography
CVD	Cardiovascular Disease
EOF	Electroosmotic Flow
EPS	Extracellular polymeric substances
FSI	Fluid Structure Interface
HA	Hyaluronic acid
MATLAB	Matrix Laboratory (numerical computing software)
MHD	Magnetohydrodynamics
MRE	Magnetic resonance elastography
MRI	Magnetic Resonance Image
ODE	Ordinary Differential Equation
OSI	Oscillatory shear index
PDE	Partial Differential Equation
RBC	Red blood cell
SA	Septic arthritis
SCD	Sickle cell disease
SD	Standard Deviation
SF	Synovial fluid

TAWSS	Time-Averaged Wall Shear Stress
WBC	White Blood Cell
WSS	Wall shear stress
sPTT	Simplified Phan-Thien/ Tanner